

Unraveling CRP/cAMP-mediated metabolic regulation in *Escherichia coli* persister cells

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eLife Assessment

The study reports an **important** finding on the role of the global metabolic regulator Crp/cAMP in the formation of antibiotic persister *Escherichia coli*. The evidence supporting the claims is **solid** including metabolomic analysis and characterization of many mutant strains.

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Abstract

A substantial gap persists in our comprehension of how bacterial metabolism undergoes rewiring during the transition to a persistent state. Also, it remains unclear which metabolic mechanisms become indispensable for persister cell survival. To address these questions, we directed our efforts towards persister cells in *Escherichia coli* that emerge during the late stationary phase. These cells have been recognized for their exceptional resilience and are commonly believed to be in a dormant state. Our results indicate that the global metabolic regulator Crp/cAMP redirects the metabolism of these antibiotic-tolerant cells from anabolism to oxidative phosphorylation. Although our data demonstrates that persisters exhibit a reduced metabolic rate compared to rapidly growing exponential-phase cells, their survival still relies on energy metabolism. Extensive genomic-level analyses of metabolomics, proteomics, and single-gene deletions consistently highlight the critical role of energy metabolism, specifically the tricarboxylic acid (TCA) cycle, electron transport chain (ETC), and ATP synthase, in sustaining persister levels within cell populations. Altogether, this study provides much-needed clarification regarding the role of energy metabolism in antibiotic tolerance and highlights the importance of using a multipronged approach at the genomic level to obtain a broader picture of the metabolic state of persister cells.

Introduction

Bacterial persisters within cell cultures constitute a small subpopulation of cells exhibiting a transient antibiotic-tolerant state (Balaban et al., 2004). While persisters have traditionally been characterized as non-growing and dormant phenotypes (Lewis, 2010), recent studies challenge these conventional hallmarks, revealing the heterogeneity of persister cells in terms of growth, metabolism, and other cellular activities (Adams et al., 2011; Arnoldini et al., 2014; Hossain et al., 2023; Ma et al., 2010; Mok, Park, et al., 2015; Orman & Brynildsen, 2013a, 2015; Shan et al., 2015; Wakamoto et al., 2013). Despite potential discrepancies in research outcomes in the field, we think that these variations arise from the intricate and diverse survival mechanisms employed by bacterial cells in response to adverse conditions, such as antibiotic treatments. Furthermore, the interplay of stochastic and deterministic factors associated with these mechanisms adds another layer of complexity, with outcomes highly contingent on factors such as cell types, antibiotics, and experimental and growth conditions (Michiels et al., 2016). The persistence phenomenon presents a significant health concern (Huemer et al., 2020; Murray et al., 2022), as the transient antibiotic-tolerant state of persister cells promotes recurrent infections (Fisher et al., 2017) and establishes them as a reservoir for the emergence of antibiotic-resistant mutants (Barrett et al., 2019; Levin-Reisman et al., 2017; Windels et al., 2019).

Drug tolerance is a widespread phenomenon observed in both prokaryotic and eukaryotic cell types. Otto Warburg's research in the early twentieth century unveiled an intriguing aspect of mammalian cell metabolism known as "aerobic glycolysis," wherein proliferating cells (e.g., tumor cells) derive energy predominantly through glycolysis, even in the presence of oxygen (Warburg, 1925). This metabolic reprogramming involves restricting entry into the TCA cycle through precise enzymatic control, diverting glycolytic intermediates towards anabolic pathways. This adaptation supports the extensive biosynthesis required for active cell proliferation in tumors (Hanahan & Weinberg, 2011). Remarkably, tumorigenic persisters that exist in a non-proliferating state may not primarily depend on aerobic glycolysis. Instead, there is substantial evidence suggesting that these cells rely on energy metabolism (Porporato et al., 2018; Vasan et al., 2020); however, the presence of this metabolic state in antibiotic-tolerant bacteria is still a question mark (Orman & Brynildsen, 2015). Reprogramming energy metabolism seems to be an evolutionarily conserved strategy for cells facing stress or adverse conditions, as these cells might benefit from the significantly higher ATP production efficiency provided by oxidative phosphorylation (Hanahan & Weinberg, 2011). The identification of a mechanism shared by diverse cell types could open avenues for the development of global strategies to target drug-tolerant cells effectively.

A recent study suggests that bacterial persisters constitute a stochastically formed subpopulation of low-energy cells, despite some observed overlap in ATP levels between antibiotic-sensitive and persister cells (Manuse et al., 2021). The persister cells examined in that study were derived from an aged stationary phase culture (48 hours post-inoculation) (Manuse et al., 2021), a condition known to elevate the number of non-growing cells, which do not promptly resume growth upon transfer to a fresh medium (Mohiuddin, Kavousi, et al., 2020). While the non-growing cells formed during the stationary phase have reduced metabolic activity compared to growing cells, as demonstrated in our earlier study (Orman & Brynildsen, 2013a), they still exhibit a certain degree of respiratory activity (Orman & Brynildsen, 2015). In a recent study measuring ATP levels in viable but non-culturable (VBNC; a phenotype that is antibiotic tolerant but unable to resume growth after antibiotic removal), persister and antibiotic-sensitive cells from 24-hour stationary phase cultures, a significant overlap in intracellular ATP concentrations was observed between antibiotic-sensitive and persisters cells (Li et al., 2024). On the other hand, VBNC cells exhibited drastically lower ATP levels compared to both persister and antibiotic-

sensitive cells (Li et al., 2024 [DOI](#)). Another independent study, which utilized single-cell analysis, reported that ofloxacin persisters were metabolically active cells in exponentially growing cultures before treatment and these cultures were obtained from 16-hour overnight precultures (not aged) (Goormaghtigh & Van Melderen, 2019 [DOI](#)). These findings suggest that while persisters formed during exponential growth may initially retain metabolic activity, they may gradually transition to reduced energy states as they age within the culture. The nature of persister-cell metabolism in bacteria is a topic that has long been a point of contention in scientific circles. The controversy surrounding this topic primarily arises from studies that rely on bacteriostatic chemicals or a limited number of gene deletions, or direct comparisons to exponentially growing cells, all of which have inherent drawbacks (Conlon et al., 2016 [DOI](#); Manuse et al., 2021 [DOI](#); Orman & Brynildsen, 2013a [DOI](#), 2015 [DOI](#)). The metabolism of persister cells, a very complex phenomenon, cannot be easily characterized by a simplistic term such as “metabolic dormancy”. Even if persister cells may exhibit a lower metabolism compared to the vast majority of rapidly growing exponential-phase cells, they may still rely on energy metabolism for their functioning (Amato et al., 2014 [DOI](#); Prax & Bertram, 2014 [DOI](#)). In fact, analyzing published studies collectively suggests that bacterial cell metabolism undergoes intricate alterations or rewiring as they transition into a tolerant state, and these alterations seem to be highly dependent on the specific conditions tested (Adams et al., 2011 [DOI](#); Arnoldini et al., 2014 [DOI](#); Ma et al., 2010 [DOI](#); Mok, Park, et al., 2015 [DOI](#); Orman & Brynildsen, 2013a [DOI](#), 2015 [DOI](#); Shan et al., 2015 [DOI](#); Wakamoto et al., 2013 [DOI](#)).

To gain a better understanding of the critical role of energy metabolism in persister cell survival, we have focused our research on antibiotic-tolerant cells formed during the stationary phase, given that these cells are known to be highly resilient and capable of surviving a variety of stressors, including antibiotics, and are assumed to be dormant (Balaban et al., 2004 [DOI](#); Orman & Brynildsen, 2015 [DOI](#)). These cells are also referred to as type I persisters, which cannot readily resume growth when diluted in fresh media during the lag phase (Balaban et al., 2004 [DOI](#)). Previous studies showed that these persister cells can metabolize specific carbon sources that make them susceptible to aminoglycosides (AG) (Allison et al., 2011 [DOI](#); Mok, Orman, et al., 2015 [DOI](#); Orman & Brynildsen, 2013b [DOI](#)). Their AG susceptibility is due to increased AG uptake, which is facilitated by increased ETC activity and membrane potential (Allison et al., 2011 [DOI](#)). The presence of active energy metabolism in antibiotic-tolerant, non-growing cells may indeed explain their rapid killing by AG in the presence of carbon sources (Orman & Brynildsen, 2013b [DOI](#), 2015 [DOI](#)). When the knockout strains of global transcriptional regulators (i.e., ArcA, Cra, Crp, DksA, Fnr, Lrp, and RpoS) were screened using the AG potentiation assay in our previous study, the results showed that the panel of carbon sources tested potentiated the AG killing of tolerant cells derived from most knockout strains, except for Δcrp and $\Delta cyaA$ (Mok, Orman, et al., 2015 [DOI](#)). This can be attributed to the lack of active energy metabolism in these mutant strains, as the Crp/cAMP potentially shapes persister cell metabolism during the stationary phase. Depletion of primary carbon sources activates adenylate cyclase (CyaA) (Pastan & Perlman, 1970 [DOI](#)), increasing cyclic-AMP (cAMP) levels in cells (Bettenbrock et al., 2007 [DOI](#); Park et al., 2006 [DOI](#)). The cAMP molecules, along with its receptor protein (Crp), activate genes related to the catabolism of secondary carbon sources, potentially supporting cellular functions and energy levels (Deutscher, 2008 [DOI](#); Fic et al., 2009 [DOI](#); Görke & Stülke, 2008 [DOI](#); Kolb et al., 1993 [DOI](#)). Here, using metabolomics, proteomics, and high-throughput screening of single-gene deletion strains, we have provided evidence that the Crp/cAMP regulatory complex maintains an active state of energy metabolism while downregulating anabolic pathways in the antibiotic-tolerant persister cells.

Results

Disruption of the Crp/cAMP complex affects the formation of persister cells at the late stationary phase

Since Crp/cAMP-mediated metabolic changes can be induced by nutrient depletion during the stationary phase, we wanted to assess the effects of deleting the *crp* and *cyaA* genes (Δcrp and $\Delta cyaA$) on both persister cell formation and metabolism during this phase. As anticipated, the deletion of the *cyaA* gene resulted in a notable reduction in intracellular cAMP concentration (**Figure 1—figure supplement 1**). However, the Δcrp strain exhibited an increase in cAMP concentration, potentially due to the negative feedback regulatory mechanism of the Crp/cAMP complex for the *cyaA* gene promoter (Keseler, 2004; Majerfeld et al., 1981) (**Figure 1—figure supplement 2**). When comparing the growth curves of *E. coli* wild-type (WT), Δcrp , and $\Delta cyaA$ main cultures under identical conditions studied here (see **Materials and Methods**), all three strains started to enter the stationary phase around 5 hours, with the mutant strains exhibiting slightly lower optical density levels than the WT at this time point (**Figure 1—figure supplement 3**). Also, our data provide evidence of an increase in cAMP levels in WT cells during their transition into the stationary phase (**Figure 1—figure supplement 4**), aligning with existing literature (Bettenbrock et al., 2007; Keseler, 2004; Park et al., 2006). For type I persister quantification, we diluted cells in the fresh medium, consistent with previous studies (Balaban et al., 2004; Orman & Brynildsen, 2015), at early (t=5 h) and late (t=24 h) stationary phases and subsequently exposed them to an extended period (20 h) of ampicillin or ofloxacin treatment (200 μ g/mL ampicillin and 5 μ g/mL ofloxacin). These treatments were carried out at concentrations surpassing the minimum inhibitory concentrations (MIC), necessary for the selection of antibiotic-tolerant persister cells (**Supplementary File 1**) (Keren, Kaldalu, et al., 2004). Also, type I persisters, formed during the stationary phase, exhibit a slow transition from a non-growing state to an active state when transferred to a fresh medium in the lag phase (Balaban et al., 2004; Jøers & Tenson, 2016; Orman & Brynildsen, 2015; Vulin et al., 2018); therefore, this transition requires longer antibiotic treatment durations. Moreover, we transferred an equal number of cells from each strain to the fresh medium to ensure consistency in cell numbers. To assess persistence, we collected samples during antibiotic treatment, washed them to remove antibiotics, and plated them on agar media to quantify colony-forming units (CFU) of surviving cells (see **Materials and Methods**). The resulting biphasic kill curves—plots of CFU levels over treatment time—are characteristic of persistence phenotypes (**Figure 1**). Notably, the WT strain showed a marked increase in both ampicillin- and ofloxacin-persister cells during the late stationary phase, in contrast to the mutant strains, where no such increase was observed (**Figure 1A, B**). However, this trend was not observed in the early stationary phase (**Figure 1A, B**). To confirm that the observed decrease in persister levels in the mutant strains in the late stationary phase is solely attributed to the perturbation of the Crp/cAMP regulatory network, we reintroduced *crp* expression to the Δcrp strain using a low-copy plasmid carrying the *crp* gene and its promoter. As a control, we utilized an empty vector of the same plasmid. The results demonstrated that the expression of *crp* restored the persister level in the mutant strain, while the plasmid itself had no impact on persister levels (**Figure 1—figure supplement 5A,B**). Altogether, these findings highlight the significant role of Crp/cAMP in ampicillin and ofloxacin persister formation in the late stationary phase.

Ampicillin and ofloxacin, both broad-spectrum antibiotics with a strong dependence on cell metabolism (E. J. Zheng et al., 2020), target cell wall synthesis and DNA gyrase activity, respectively. In addition to these two antibiotics, we also quantified aminoglycoside-persister levels in both WT and mutant strains. This was achieved by exposing diluted cells from both early and late stationary phases to 50 μ g/mL gentamicin, a concentration exceeding the MIC levels (**Supplementary File 1**). After prolonged gentamicin exposure (20 h), the tolerant cell colonies were found to be below the limit of detection for all strains and conditions (**Figure 1C**, and **Figure 1—figure supplement 5C**). Although bacterial tolerance can vary significantly depending on the specific antibiotics and growth phase used (Hofsteenge et al., 2013), this outcome contrasts starkly with the observed levels of ampicillin and ofloxacin persisters in WT

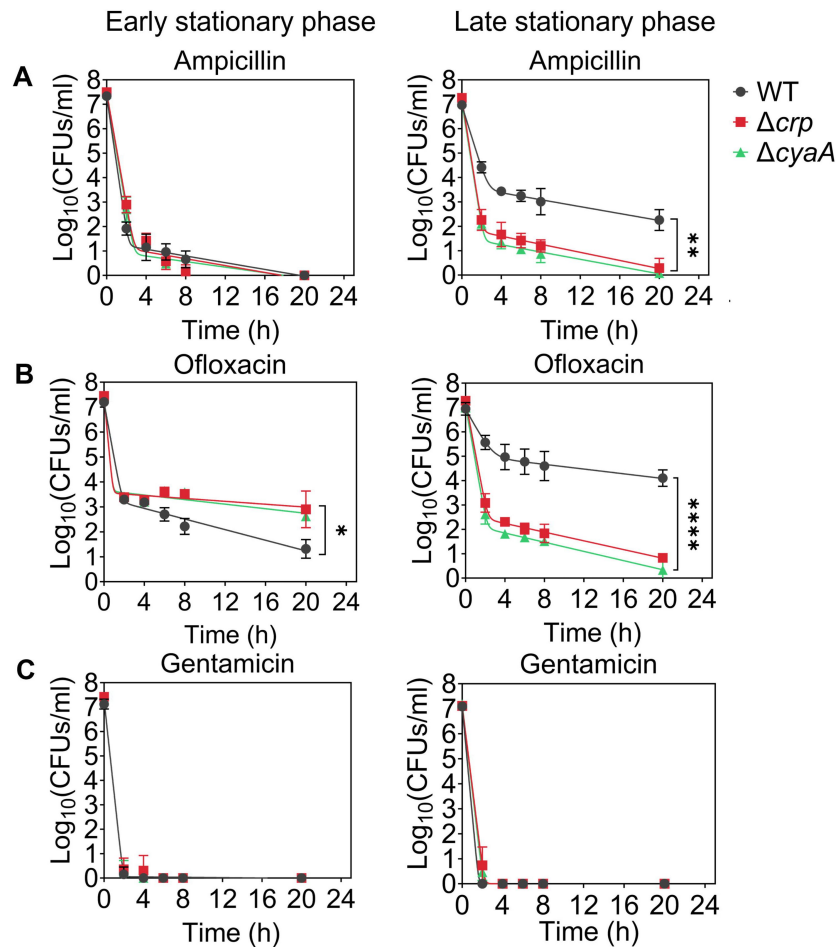


Figure 1

Crp/cAMP Regulation of Persister Cell Formation in the Stationary Phase.

E. coli K-12 MG1655 WT and mutant cells at early ($t=5$ h) and late ($t=24$ h) stationary phases were transferred to fresh medium with antibiotics for persister cell quantification. At time points 0, 2, 6, 8, and 20 h, 1 mL of the treated culture was washed with 1X phosphate-buffered saline (PBS) to remove antibiotics. It was then serially diluted and plated on an agar plate to count the colony-forming units (CFUs). **(A)** Persister levels of ampicillin-treated culture with an antibiotic concentration of 200 $\mu\text{g/mL}$. **(B)** Persister levels of ofloxacin-treated culture with an antibiotic concentration of 5 $\mu\text{g/mL}$. **(C)** Persister levels of gentamicin-treated culture with an antibiotic concentration of 50 $\mu\text{g/mL}$. The number of biological replicates is $n=4$ for all panels. Biphasic kill curves were generated using a non-linear model (see **Materials and Methods**). Statistical significance tests were conducted using F-statistics (* $P < 0.05$, ** $P < 0.01$, **** $P < 0.0001$). The data for each time point represent the mean value $\pm$ standard deviation.

(**Figure 1**). The mechanism by which aminoglycosides eliminate persister cells in the WT strain may be linked to their metabolism, given that aminoglycoside uptake is an energy-requiring process (Allison et al., 2011; Taber et al., 1987), and this aspect will be explored further in the subsequent section.

We would like to highlight that antibiotic concentrations, washing procedures (to remove antibiotics), and agar plate incubation times for CFU enumeration may affect experimental outcomes. However, variations in these parameters, including lower antibiotic concentrations normalized to the MIC of each strain (5× or 10× MIC; see **Figure 1—figure supplement 6**), additional washing steps to ensure complete removal of antibiotics (see **Figure 1—figure supplement 6**), and extended agar plate incubation times of up to 48 hours (see **Figure 1—figure supplement 7**), did not affect the persistence phenotype. The Δcrp and $\Delta cyaA$ strains consistently showed reduced ampicillin and ofloxacin persistence compared to WT, whereas all three strains showed no detectable persisters following gentamicin treatment (**Figure 1—figure supplement 6, 7**). Moreover, we examined the impact of Crp/cAMP disruption in the HipA7 strain to assess whether the role of Crp/cAMP in persistence extends to a genetically sensitized, persister-enriched background. Deletion of either *crp* or *cyaA* in the HipA7 background significantly reduced persistence to both ampicillin and ofloxacin during the late stationary phase (**Figure 1—figure supplement 8**), mirroring the effects observed in the WT strain. Altogether, these results show that disruption of the Crp/cAMP regulatory network impairs persister formation during the late stationary phase, underscoring its critical role in ampicillin and ofloxacin tolerance.

Crp/cAMP complex governs *E. coli* stationary phase metabolism

To determine whether the reduced ampicillin and ofloxacin persister levels at the late stationary phase in the mutant strains (**Figure 1A, B**) are linked to stationary phase metabolism, we utilized untargeted mass spectrometry (MS). This approach facilitated the quantification of metabolites in Δcrp cells, allowing for a comparison with WT controls in both the early and late stationary phases (**Figure 2A**). Since both Δcrp and $\Delta cyaA$ strains exhibit the same persistence phenotype, and *crp* deletion effectively abolishes Crp/cAMP complex function, we used the Δcrp mutant for metabolomic profiling to capture key regulatory changes while maintaining experimental feasibility. The metabolomics data were subjected to unsupervised hierarchical clustering, and metabolites identified in independent biological replicates of each strain and condition were found to cluster together (**Figure 2A**), thus confirming the reproducibility of our data.

To elucidate the upregulated and downregulated metabolic pathways in the mutant strain as compared to the WT strain, we performed enrichment analyses utilizing MetaboAnalyst (Lu et al., 2023). For downregulated pathways, we considered a threshold ratio of 0.5 or lower, where the ratio indicates metabolite levels in the mutant strain relative to the WT (Supplementary File 2A). Conversely, for upregulated pathways, the threshold ratio was set at 2 or higher (Supplementary File 2B). The enrichment ratio for each pathway was calculated based on the number of metabolite hits compared to the expected hits derived from the chemical structure library (Lu et al., 2023). Our extensive comparison of the mutant cells to the WT cells through pathway enrichment analysis (refer to **Figure 2B** and **Figure 2—figure supplements 1–3** for pairwise comparison of different conditions) revealed several important findings:

- (i) During the early stationary phase, we observe a slight downregulation in the abundance of TCA cycle metabolites, including citrate and fumarate, in the Δcrp strain compared to WT (**Figure 2C** and **Figure 2—figure supplement 1**). However, as the late stationary phase progresses, the downregulation in both TCA cycle and pentose phosphate metabolism becomes more pronounced in the Δcrp strain (**Figure 2B, D**).

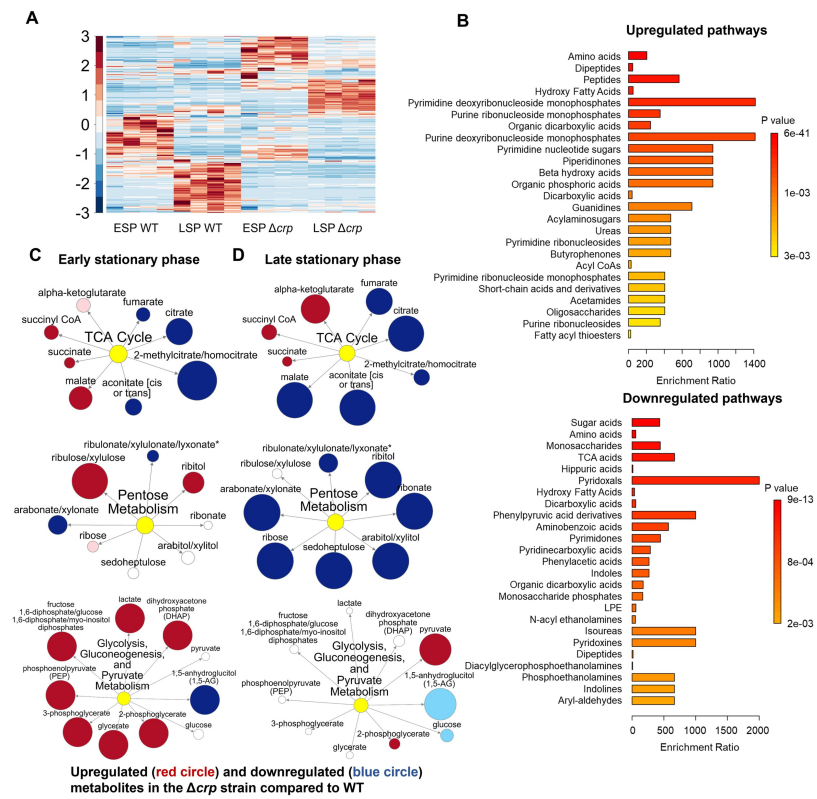


Figure 2

The effect of Crp/cAMP on Persister Cell Metabolism during Stationary Phase.

(A) MS analysis of *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyoA$ at early (t=5 h) and late (t=24 h) stationary phases. Unsupervised hierarchical clustering was applied to standardized metabolic data. Each column represents a biological replicate. n=4. (B) Pathway enrichment analysis was conducted using MetaboAnalyst (Lu et al., 2023). Upregulated and downregulated pathways of the Δcrp strain compared to WT in the late stationary growth phase were provided in this figure. (C, D) Pathway enrichment maps comparing metabolites of the TCA cycle, pentose phosphate metabolism, glycolysis, gluconeogenesis, and pyruvate metabolism in Δcrp versus WT for early and late stationary phase conditions, respectively. Circle size corresponds to the ratio of normalized metabolite intensities between mutant and control cells. Blue ($P \leq 0.05$ for dark blue; $0.05 < P < 0.10$ for light blue) and red ($P \leq 0.05$ for dark red; $0.05 < P < 0.10$ for light red) indicate significantly downregulated or upregulated metabolites in the mutant compared to the control. White signifies no significant difference. n=4 for all panels. ESP: Early stationary phase, LSP: Late stationary phase.

- **(ii)** The Δcrp strain exhibits upregulation of several metabolites compared to WT during both early and late stationary phases, primarily associated with anabolic pathways. Particularly, the upregulation of some of these pathways becomes more pronounced during the late stationary phase. These pathways include crucial metabolites like deoxyribonucleosides and ribonucleosides (which play essential roles in DNA and RNA synthesis), fatty acids and carboxylic acids (the main components of bacterial cell membranes), and peptides (which are linked to protein synthesis) (**Figure 2B** [↗](#)).
- **(iii)** During the early stationary phase, we noticed a significant upregulation in the abundance of intermediate metabolites related to glycolysis, gluconeogenesis, and pyruvate metabolism in mutant cells compared to WT cells (**Figure 2C** [↗](#)). This observation is not surprising, as the inhibition of the TCA cycle in the mutant strain could potentially redirect metabolic fluxes toward glycolysis and lactate metabolism.

Altogether, our metabolic data indicate that, in the stationary phase, WT cells maintain their energy metabolism to some extent while downregulating their anabolic pathways (**Figure 2A** [↗](#)). This metabolic state appears to be regulated by the Crp/cAMP complex, as perturbing its function leads to a significant downregulation of energy metabolism and an upregulation in the abundance of anabolic metabolites (**Figure 2B** [↗](#)).

Proteomics analysis revealed upregulated pathways in the Δcrp strain associated with anabolic metabolism, alongside the downregulation of key proteins in energy metabolism

Since the Crp/cAMP complex acts as a transcriptional regulator affecting the expression of metabolic proteins whose abundance directly affects cellular metabolites, we performed untargeted proteomics, our second genomic-level study, to further validate our results. Considering the noticeable metabolic alterations observed during the late stationary phase, we utilized MS to quantify proteins in Δcrp cells and compared them to WT controls at this stage. The resulting proteomics data were subjected to unsupervised hierarchical clustering, and the proteins identified in independent biological replicates of each strain were found to cluster together, confirming the consistency of our findings (**Figure 3—figure supplement 1** [↗](#)). By analyzing protein-protein association networks and employing functional enrichment through STRING (Szklarczyk et al., 2021 [↗](#), 2023 [↗](#)), which integrates various functional pathway classification frameworks such as Gene Ontology annotations, KEGG pathways, and UniProt keywords, we pinpointed various upregulated and downregulated pathways in the mutant strain when compared to the WT (Supplementary File 3). The upregulated pathways are associated with anabolic metabolism and encompass peptidoglycan metabolic processes, cell wall organization or biogenesis, cellular component organization or biogenesis, regulation of cell shape, cell cycle, and cell division (**Figure 3A** [↗](#)). Also, the mutant strain displayed downregulated pathways, encompassing glycerol metabolism, TCA cycle, pyruvate metabolism, glycolysis, and various pathways associated with ribosome and transcriptional factor activity (**Figure 3B** [↗](#)). Our analysis specifically pinpointed a cluster of proteins involved in energy metabolism and respiratory processes. Notably, this cluster includes GltA, SdhB, SucC, SucD, FrdB, FrdA, AcnA, AceA, and Mdh proteins, which play crucial roles in either the TCA cycle or as membrane-bound components of ETC (**Figure 3B** [↗](#)). Altogether, the alignment between metabolomics and proteomics analyses provides additional validation for the Crp/cAMP-mediated metabolic state.

Crp/cAMP complex shapes *E. coli* cell proliferation dynamics

The omics data suggest that the upregulation in the abundance of anabolic metabolites and proteins, particularly those related to cell wall organization or biogenesis, cell cycle, and cell division, in the stationary-phase mutant cells, would likely enhance their ability to resume growth upon transitioning to fresh medium. To investigate this, we utilized a cell proliferation assay that

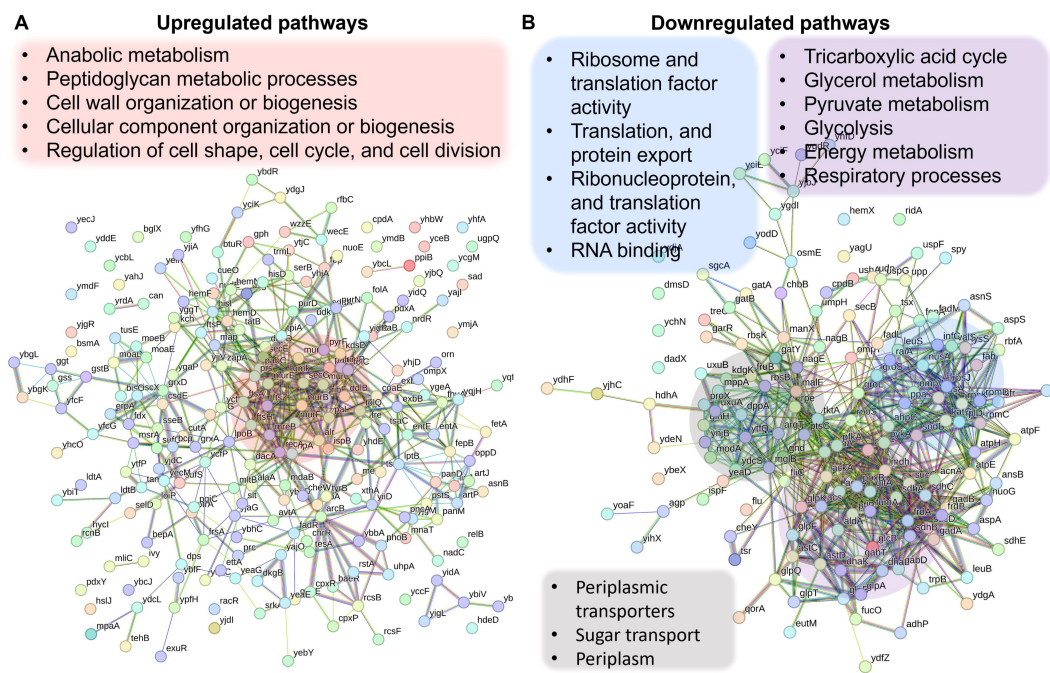


Figure 3

Validation of Crp/cAMP-Mediated Metabolic State in Persister Cells Through Proteomics Analysis.

Pathway enrichment analysis was conducted in STRING (Szklarczyk et al., 2021 [\[1\]](#), 2023 [\[2\]](#)) for upregulated (**panel A**) and downregulated (**panel B**) proteins. Genes highlighted in red are linked with the upregulated protein networks, while genes in blue, gray, and purple correspond to those in the downregulated protein network. The visual network in STRING illustrates protein interactions. In evidence mode, color in the network represents the interaction evidence of data support, derived from curated databases, experimental data, gene neighborhood, gene fusions, co-occurrence, co-expression, protein homology, and text mining (Szklarczyk et al., 2021 [\[1\]](#), 2023 [\[2\]](#)).

employed an inducible fluorescent protein (mCherry) expression cassette. This assay provided us with the ability to monitor non-growing cells at a single-cell resolution, as described previously (Orman & Brynildsen, 2013a [↗](#); Roostalu et al., 2008 [↗](#)). The mCherry expression cassette is controlled by an isopropyl β-D-1-thiogalactopyranoside (IPTG) inducible synthetic T5 promoter that was previously inserted into the chromosome of an *E. coli* strain carrying a *lacI^q* promoter mutation (Orman & Brynildsen, 2013a [↗](#)). This configuration allowed for precise regulation of mCherry expression using IPTG. Here, we introduced *crp* and *cyaA* deletions into this strain. These deletions reduced the persistence of the mCherry-expressing *E. coli* strain in the late stationary phase (**Figure 4—figure supplement 1** [↗](#)), consistent with the findings presented in **Figure 1** [↗](#). To perform the growth assay, we induced mCherry expression in the main cultures and then washed the cells to remove IPTG. The cells were then inoculated into a fresh medium without IPTG and their growth was analyzed with a flow cytometer. This allowed us to track the dilution of mCherry protein within the cells, which served as an indicator of cell proliferation. As shown in **Figure 4A** [↗](#), initially, all cells exhibited high red fluorescence. However, as cells underwent division, the red fluorescence of the overall population decreased in the absence of the inducer (**Figure 4A** [↗](#)). Notably, within the WT strain, a subpopulation from the late stationary phase cultures displayed constant fluorescence levels, indicating their inability to divide (**Figure 4B** [↗](#)). In contrast to the WT strain, we did not detect similar subpopulations in the mutant strains (**Figure 4B** [↗](#)). Additionally, this subpopulation of non-growing cells does not emerge during the early stationary phase cultures (**Figure 4A** [↗](#)). This observation provides an explanation for the observed reduction in persister levels in these mutant strains in the late stationary phase, as the enrichment of persister cells within these non-growing cell subpopulations was reported in previous studies (Joers et al., 2010 [↗](#); Orman & Brynildsen, 2013b [↗](#); Roostalu et al., 2008 [↗](#)).

To confirm the significance of Crp/cAMP in the formation of non-growing cells, we introduced the expression plasmid carrying the *crp* gene into the Δcrp strain. As anticipated, the introduction of the *crp* expression plasmid resulted in the emergence of a non-growing population within the culture, contrasting with the mutant strain containing the empty plasmid vector used as a control (**Figure 4—figure supplement 2A, B** [↗](#)). The reduced capacity of stationary-phase WT cells to initiate proliferation upon transfer to a fresh medium suggests the possible presence of an extended lag phase in these cells. To investigate this, we employed flow cytometry to precisely quantify cell numbers and generate growth curves for both WT and mutant strains. As anticipated, the growth curve of the WT strain displayed a slower initial growth rate and a prolonged lag phase duration compared to the mutant strains (**Figure 4C** [↗](#)). Conversely, the mutant strains displayed a shorter lag phase, yet they demonstrated an increased doubling time in the exponential phase compared to WT (**Figure 4C** [↗](#)), which was also anticipated, considering their decreased reliance on oxidative phosphorylation due to TCA cycle inhibition. Altogether, these results provide additional support and validation for the findings from our metabolomics and proteomics data, as our data reveals a correlation between the abundance of molecules associated with cell division and the ability of the stationary phase cells to resume growth.

Persister cells rely on energy metabolism

The Crp/cAMP-mediated metabolic state, characterized by increased respiration in WT compared to mutant strains, was further validated using redox sensor green (RSG) dye and a reporter plasmid measuring the promoter activity of succinate:quinone oxidoreductase (SQR) genes. The SQR reporter system (Zaslaver et al., 2006 [↗](#)) employs green fluorescent protein (GFP) expression, regulated by the promoter of the SQR operon, which includes the *sdhA*, *sdhB*, *sdhC*, and *sdhD* subunit genes. The SQR complex plays a vital role in cellular metabolism by catalyzing the oxidation of succinate to fumarate concurrently with the reduction of ubiquinone to ubiquinol, thus directly linking the TCA cycle with the respiratory ETC (Keseler, 2004 [↗](#)). Our results indicate an upregulation of the SQR promoter activity in WT cells compared to the mutant strains in the stationary phase, validating the findings from our metabolomics and proteomics data (**Figure 5A** [↗](#)). The RSG dye, on the other hand, serves as a well-established metabolic indicator, measuring bacterial oxidation-reduction activity, a crucial function involving the ETC driven by

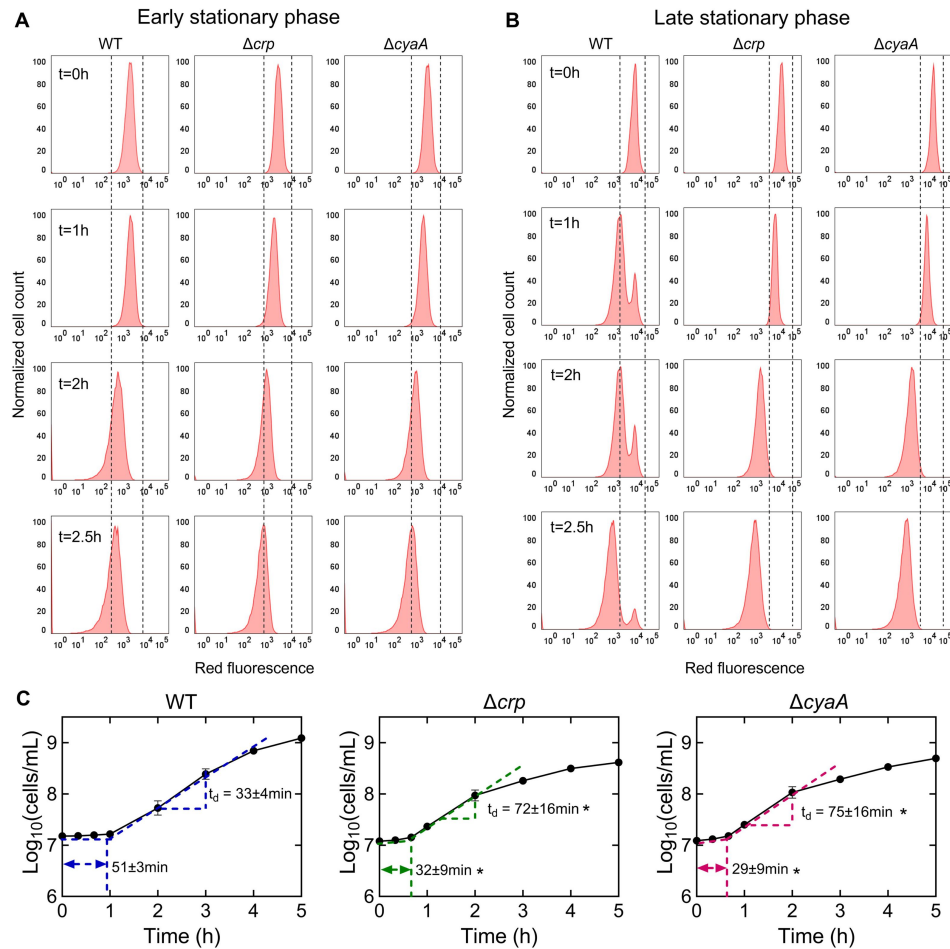


Figure 4

The Role of Crp/cAMP in Non-Growing Cell Formation.

(A, B) Flow cytometry histograms depict mCherry expression in *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$ at early ($t=5$ h) and late ($t=24$ h) stationary phases, respectively. Cells containing an IPTG-inducible mCherry expression system were cultivated with IPTG. After washing and dilution of early and late stationary phase cells in IPTG-free fresh media, fluorescence was tracked in non-growing and growing cells for 2.5 hours. The panel is a representative biological replicate. Consistent results were seen across all 3 biological replicates. (C) Growth curves of WT, Δcrp , and $\Delta cyaA$ cultures were determined using flow cytometry to calculate lag and doubling times. Lag times were calculated using the "Microbial lag phase duration calculator" (Opalek et al., 2022). Doubling times were computed using the formula $t_d = \Delta t / (3.3 \times \log_{10}(N/N_0))$, $n=3$. *Statistical significance observed between control and mutant strains ($P < 0.05$, 2-tailed t-test). The data for each time point represent the mean value $\pm$ standard deviation.

the TCA cycle. Once reduced by bacterial reductases, the RSG dye emits a stable green fluorescent signal (**Figure 5—figure supplement 1**). Our data demonstrate that the redox activities of WT cells are much higher and more heterogeneous compared to those of the mutant strains in the late stationary phase, further corroborating the results from our preceding analyses (**Figure 5B**). Furthermore, we utilized a methodology that integrates the mCherry expression system, flow cytometry, and ampicillin-mediated cell lysis, to determine whether persister cells in WT still maintain their respiration. In this assay, both the WT and mutant strains carrying the mCherry expression system were exposed to ampicillin after transferring them to a fresh medium. The inducer was added to both growth and treatment cultures to sustain the cells' red signals. Unlike other antibiotics, ampicillin disrupts cell wall synthesis, leading to the lysis of cells upon their resumption of growth. As seen in **Figure 5—figure supplement 2**, the cells that were lysed lost their mCherry signals. On the other hand, the resilient, tolerant cells that evaded ampicillin-induced lysis maintained their mCherry levels throughout the treatment. In the mutant strains, ampicillin was effective in lysing almost all cells as anticipated, leaving no or small number of intact cells (**Figure 5—figure supplement 2**). However, in the WT strain, we detected a subpopulation of intact cells throughout the entire treatment period (**Figure 5C**, and **Figure 5—figure supplement 2**). While the population-level redox activities of these tolerant intact cells in WT are lower than those of exponential phase cells, they still displayed a significant increase in RSG levels compared to cell populations before antibiotic treatments or untreated control cells subjected to identical conditions (**Figure 5D**), suggesting that they maintain steady-state energy metabolism. We want to highlight that not all intact cells in the WT strain reported here are persisters. A significant portion comprises 'viable but non-culturable' (VBNC) cells, and WT cells exhibit markedly higher VBNC levels than Δcrp and $\Delta cyaA$ strains (**Figure 5—figure supplement 3**). VBNC cells can be quantified from intact cells following beta-lactam treatments (Orman & Brynildsen, 2013b; Roostalu et al., 2008). These cells may exhibit metabolic activities but are unable to readily colonize upon transfer to fresh medium (Arapetyan et al., 2018; Orman & Brynildsen, 2013b). Collectively, our metabolic measurement data (**Figure 5A-C**) aligns with the findings from our omics analyses, and the number of intact cells observed in WT after beta-lactam treatment is consistent with the count of non-growing cells in WT (**Figure 4B** vs **Figure 5—figure supplement 2**). These non-growing cells are anticipated to be less susceptible to lysis by beta-lactams (Orman & Brynildsen, 2013b; Roostalu et al., 2008).

The genomic-level screening of *E. coli* knockout underscores the significance of energy metabolism in sustaining the viability of persister cells

Although some metabolic genes, including those encoding the TCA cycle (e.g., *sdhA*, *sucB*, *mdh*, *icd*), have been studied in *E. coli* (Luidalepp et al., 2011; Ma et al., 2010; Manuse et al., 2021; Orman & Brynildsen, 2015; Yu et al., 2019), a comprehensive genomic-level screening strategy is necessary to validate which metabolic pathways are truly associated with antibiotic tolerance. To further underscore the importance of energy metabolism, we conducted a high-throughput screening of 149 different *E. coli* K-12 BW25113 mutant strains from the Keio knockout library (Baba et al., 2006). The selected strains are related to central carbon metabolism, encompassing glycolysis, pentose phosphate pathways, TCA cycle, ETC, ATP synthase, and fermentation pathways (**Figure 5E**). While the deletion of genes related to glycolysis and pentose phosphate pathways did not affect antibiotic tolerance in the cells, some mutant strains associated with cytochrome bo and quinone oxidoreductase complexes (e.g., *cyo* genes, and *nuoL*) exhibited enhanced tolerance (**Supplementary File 4**). However, the mutant strains exhibited reduced tolerance to both antibiotics compared to the control *E. coli* K-12 BW25113 WT strain were found to be largely associated with the TCA pathway (*sucA*, *sucB*, *lpd*, *sucC*, *sucD*, *sdhA*, *sdhB*, *sdhC*, *sdhD*, *gltA*, *acnB*, *aceE*, *fumA*, *mdh* and *fumC*), ETC (*nuoB*, *nuoC*, *nuoI*, *nuoK*, *nuoM*, and *narV*), ATP synthesis (*atpA*, *atpB*, *atpC*, *atpD*, *atpE*, and *atpH*), and mixed acid fermentation pathways (*ldhA*, *fdhF*, *pta*, *adhE*, and *frdC*) (**Figure 5E**, and **Supplementary File 4**). We acknowledge that the Keio strains were

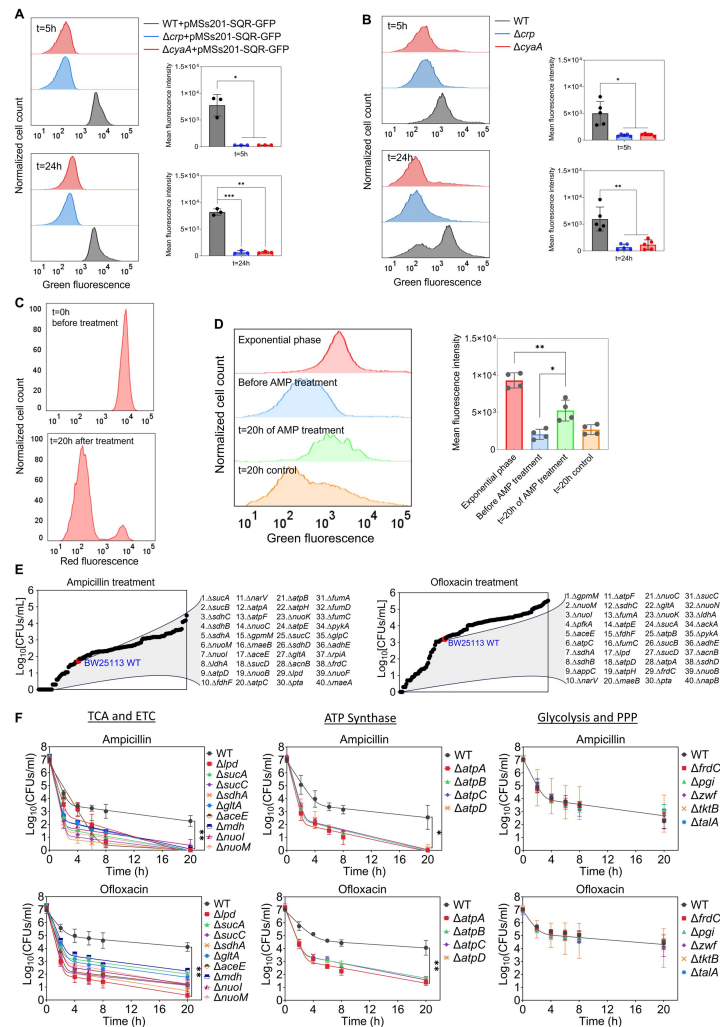


Figure 5

Crp/cAMP-Mediated Metabolic State of Persister Cells.

(A) GFP reporter plasmid introduced into *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$ cells to monitor SQR gene activity. Flow cytometry was used to detect activity at early (t=5 h) and late (t=24 h) stationary phases. The panel on the left represents a biological replicate, and the results are consistent across all 3 replicates, as demonstrated in the panel on the right. Statistical significance observed between control and mutant groups (* $P < 0.05$, ** $P < 0.01$, *** $P < 0.001$, 2-tailed t-test). **(B)** Redox activities of *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$ cells were measured at early (t=5 h) and late (t=24 h) stationary phases by flow cytometry using a RSG dye. This dye fluoresces green after reduction by bacterial reductases. A representative biological replicate is shown (left), with consistent results across all 5 replicates (right). Statistical significance observed between control and mutant groups (* $P < 0.05$, ** $P < 0.01$, 2-tailed t-test). **(C)** *E. coli* cells with integrated mCherry expression system used to validate cellular respiration. Cells were diluted into fresh media and treated with ampicillin (200 μ g/mL) for 20 hours. Flow cytometry measured the red fluorescence of intact surviving cells. A representative biological replicate is shown, with consistent results across all 3 replicates. **(D)** RSG levels of cells (carrying the mCherry expression system) at exponential phase (t=3 h); cells before ampicillin treatment; non-lysed (intact) cells after 20 hours of ampicillin treatment; and untreated cells after 20 hours of culturing. A representative biological replicate is shown (left), with consistent results across all 4 replicates (right). Statistical significance observed between intact antibiotic-treated cells and others (* $P < 0.05$, ** $P < 0.01$, 2-tailed t-test). **(E)** High-throughput screening of mutants from the Keio collection. The mutant strains selected are associated with central metabolism. Stationary phase cells were diluted 100-fold in fresh medium and treated with ampicillin (200 μ g/mL) or ofloxacin (5 μ g/mL) for 20 hours. Treated cultures were washed, serially diluted, and plated on agar plates to quantify CFUs. **(F)** Genes related to the TCA cycle, ETC, ATP synthase, glycolysis, and pentose phosphate pathway (PPP) were knocked out and then treated with ampicillin (200 μ g/mL) or ofloxacin (5 μ g/mL) to enumerate CFUs. $n=4$. Biphasic kill curves were generated using a non-linear model. Statistical significance tests were conducted using F-statistics (* $P < 0.05$, and ** $P < 0.01$). Each data point represents the mean value $\pm$ standard deviation.

generated in a high-throughput manner, and there might be unknown errors in their genomic DNA. To ensure the reproducibility of our findings, we generated knockout strains for key genes associated with the TCA, ETC, ATP synthase, glycolysis, and pentose phosphate pathway (**Figure 5F** [↗](#), see Supplementary File 5 for detailed description of genes). We then tested their antibiotic tolerance, and our results were consistent with the omics data and screening outcomes (**Figure 5F** [↗](#)), confirming the critical role of energy metabolism, specifically the TCA cycle, ETC, and ATP synthase, in bacterial persistence. We note that equal numbers of cells from each strain were transferred into antibiotic treatment media to ensure consistency in cell numbers, and no significant growth deficiencies or differences in cell density were observed in the late stationary phase of the knockout strains (**Figure 5—figure supplement 4** [↗](#)).

Discussion

Our study highlights the crucial role of the Crp/cAMP complex in maintaining the metabolic state of stationary-phase persister cells, enabling their survival under adverse conditions. Through metabolomics, proteomics, and high-throughput screening of single-gene deletion strains, we substantiated that the Crp/cAMP regulatory complex sustains an active respiratory state while downregulating anabolic pathways in persister cells. This respiratory state is vital for the survival of persister cells, as perturbing the Crp/cAMP complex or respiration significantly reduced persister phenotypes, which may explain the previously reported decrease in antibiotic tolerance in *E. coli* cells cultured under anaerobic conditions (Orman & Brynildsen, 2015 [↗](#)). Notably, we observed an upregulation of anabolic metabolites and proteins when the Crp/cAMP regulatory complex was perturbed, particularly those associated with cell wall organization, cell cycle, and cell division, enhancing the ability of stationary-phase mutant cells to resume growth. Although the literature has shown associations between antibiotic tolerance with proteins involved in cell division and the TCA cycle (e.g., SdhA, SucB, Mdh) (Luidalepp et al., 2011 [↗](#); Ma et al., 2010 [↗](#); Orman & Brynildsen, 2015 [↗](#); Yu et al., 2019 [↗](#)), our study establishes a strong link between these critical cellular processes and the Crp/cAMP complex, providing much-needed clarity in the field. In fact, upon investigating Crp/cAMP regulons via the Ecocyc database (Keseler, 2004 [↗](#)), we identified that certain metabolic genes, deleted in our *E. coli* K-12 MG1655 background, are potentially regulated by Crp/cAMP (**Supplementary File 5**), providing additional support for the validity of our omics results.

We acknowledge that our metabolomic and proteomic data were obtained at the whole-population level, rather than from isolated persister cells. Fluorescent reporters combined with fluorescence-activated cell sorting (FACS) have been utilized to study persister cells, including in our previous studies (Amato et al., 2013 [↗](#); Orman & Brynildsen, 2013a [↗](#), 2015 [↗](#)). However, this approach only enriches for persisters rather than isolating a pure population, as persisters still constitute a small fraction of the sorted cells. Despite these limitations, our population-level analyses provide valuable insights into the role of the Crp/cAMP complex in regulating non-growing cell formation and persistence (**Figure 4** [↗](#)). We further validated these findings through single-gene deletions and flow cytometry-based assays (**Figure 5** [↗](#)), confirming a consistent reduction in both non-growing and persister cells in Crp/cAMP-deficient strains.

The deletion of *cyaA* resulted in a reduction of cAMP levels, as expected given the role of the CyaA enzyme in cAMP synthesis (**Figure 1—figure supplement 1** [↗](#)). Conversely, the removal of *crp* led to an increase in cAMP levels compared to those in wild-type cells. Notably, this increase is statistically significant (**Figure 1—figure supplement 1** [↗](#)). Although this is an interesting observation, it is likely due to the feedback regulation of the Crp/cAMP complex on *cyaA* expression (Aiba, 1985 [↗](#); Keseler et al., 2011 [↗](#); Majerfeld et al., 1981 [↗](#)). Specifically, disruption of Crp function is expected to derepress the *cyaA* promoter (P_{cyaA}), leading to increased CyaA expression and elevated cAMP levels. To test this, we employed a P_{cyaA} -*gfp* reporter plasmid (pMSs201) in *E. coli* K-12 MG1655 WT, Δ *crp*, and Δ *cyaA* strains to monitor promoter activity. As

anticipated, deletion of *crp* led to enhanced *gfp* expression, confirming that Crp/cAMP negatively regulates *cyaA* expression via feedback. These results clarify the observed cAMP differences and support the regulatory role of the Crp/cAMP complex (**Figure 1—figure supplement 2** [↗](#)).

Our findings reveal substantial heterogeneity in metabolism (measured by RSG) among WT stationary phase cells, contrasting with the more uniform behavior observed in Δcrp and $\Delta cyaA$ strains (**Figure 5B** [↗](#)). We also demonstrated the presence of two distinct populations in WT cells during the late stationary phase: one that resumes rapid growth and another subpopulation that does not resume growth when transferred to a fresh medium (**Figure 4B** [↗](#)). The absence of this non-growing cell subpopulation in the mutant strains could account for their sensitivity to aminoglycosides. However, the mechanism by which aminoglycosides kill these non-growing cells in the WT strain remains perplexing. The underlying reasons might be linked to their metabolism, as aminoglycoside uptake is an energy-requiring process, relying on the electron flow through membrane-bound respiratory chains (Allison et al., 2011 [↗](#)). Moreover, persister cells obtained from various antibiotics, such as ampicillin and ofloxacin, in WT *E. coli* were previously found to exhibit sensitivity to aminoglycosides when sugar molecules were introduced into the cultures (Allison et al., 2011 [↗](#); Orman & Brynildsen, 2013b [↗](#)). However, the enhanced sensitivity mediated by sugar molecules was reversed to its original state in a subsequent study when the Crp/cAMP complex was genetically perturbed (Mok, Orman, et al., 2015 [↗](#)). This can be attributed to the lack of active energy metabolism in these genetically altered strains, as suggested by our comprehensive genomic-level analyses here. While the absence of cell division in non-growing cell subpopulations in WT may suggest a downregulation in anabolic metabolism, their energy metabolism may remain partially active, which could potentially explain the phenomenon of aminoglycoside potentiation. Indeed, our results presented in **Figure 5D** [↗](#) support this interpretation.

Antibiotics are generally effective against proliferating bacteria, leading to the notions that tolerance is linked to temporary growth suppression and that persister cells are dormant phenotypes with repressed metabolism. Although persister cells are generally considered to have reduced metabolic activity compared to exponentially growing cells, their survival may still depend on energy metabolism. Also, the direct comparison of persister cell metabolism to that of exponentially growing cells may not be the best approach. Growing cells have a very high energy output and consume metabolites at a fast pace. Therefore, any comparison between tolerant and non-tolerant cell populations requires proper normalization techniques such as adjusting cellular metabolic activities to the amount of substrate utilized by cells. An example of this normalization was conducted by Heinemann's group (Radzikowski et al., 2016 [↗](#)), demonstrating that ATP production rates per substrate in tolerant cells exceed those of exponentially growing cells.

We diluted cells in fresh medium at early and late stationary phases before antibiotic treatments. This step is essential for quantifying type I persisters, as these cells do not readily resume growth upon dilution in the fresh medium during the lag phase (Balaban et al., 2004 [↗](#); Orman & Brynildsen, 2015 [↗](#)). We acknowledge that antibiotic tolerance is influenced by various factors, including culture dilutions, media, specific strains, antibiotics tested, treatment durations, and the growth phase during treatment administration. These factors may contribute to variations in reported persister levels observed in the $\Delta cyaA$ strain during the exponential phase (Chu et al., 2012 [↗](#); Molina-Quiroz et al., 2018 [↗](#); Parsons et al., 2024 [↗](#); Sulaiman & Lam, 2020 [↗](#); Yamasaki et al., 2020 [↗](#); A. Zeng et al., 2022 [↗](#)). To assess how *cyaA* and *crp* deletions affect antibiotic responses under conditions similar to those used by Zeng et al. (J. Zeng et al., 2022 [↗](#)) — specifically, exponential-phase *E. coli* BW25113 strains (Keio collection), lower antibiotic concentrations, and short treatments (e.g., 1 hour)—we first tested *E. coli* MG1655 WT, Δcrp , and $\Delta cyaA$ strains in late stationary phase using reduced antibiotic concentrations and shorter exposures. Both knockouts showed decreased survival following ampicillin and ofloxacin treatment compared to WT (Supplementary File 6), consistent with our findings in **Figure 1** [↗](#). In exponential-phase, the knockout strains exhibited reduced survival after ampicillin treatment but

increased survival after ofloxacin treatment relative to WT (Supplementary File 7A), again mirroring the trends in **Figure 1**. Gentamicin treatment, however, produced variable results in MG1655 knockouts, likely due to the brief 1-hour exposure being insufficient for robust conclusions (Supplementary File 7A). Notably, when we tested the corresponding Keio knockout strains in the BW25113 background, we observed increased tolerance in exponential-phase cells, reproducing Zeng et al.'s findings under their specific conditions (Supplementary File 7B), although BW25113 and MG1655 exhibited distinct persister phenotypes in exponential phase (Supplementary File 7A, B). These results, altogether, highlight the sensitivity of antibiotic tolerance and persistence phenotypes to factors such as strain background, antibiotic concentration, and treatment duration.

While we did not focus on the exponential growth phase in our study, it is noteworthy that the reduced growth rate during this phase in the mutant strains (**Figure 4C**) may explain the antibiotic tolerance observed in previous studies involving the *cyaA* deletion (Chu et al., 2012; Molina-Quiroz et al., 2018; Sulaiman & Lam, 2020; Yamasaki et al., 2020). The growth disparity noted between mutant and WT strains, particularly evident around 5 hours in our results (**Figure 1—figure supplement 3**), may also be linked to the ofloxacin persisters observed in the mutant strains at this specific time point (**Figure 1**). While slow cell growth may indeed correlate with bacterial persistence (Kaldalu & Tenson, 2019), it is important to note that the persistence associated with perturbations of metabolic genes cannot be solely attributed to the slow growth. In fact, the persistence of these mutant strains should depend on many factors (see Supplementary File 5) as reported by diverse research groups (Kim et al., 2016; Luidalepp et al., 2011; Ma et al., 2010; Orman & Brynildsen, 2015; Pandey et al., 2021; Shan et al., 2015; Spoering et al., 2006; Wang et al., 2018; Yu et al., 2019; Zalis et al., 2019). For instance, in *E. coli*, TCA inactivation was shown to decrease ampicillin and ofloxacin persistence during the lag phase (Orman & Brynildsen, 2015), yet it enhances gentamicin tolerance in the exponential phase, which remains unexplained by factors such as cell growth, redox activities, proton motive force (PMF), or ATP levels (Shiraliyev & Orman, 2023). Furthermore, gene deletions often trigger pleiotropic effects, leading to unique tolerance mechanisms not evident in wild-type strains. In our Keio screening data analysis, we observed that the deletion of *icd* appeared to enhance persistence (Supplementary File 4), in line with a previous study (Manuse et al., 2021). The *icd* gene encodes a TCA cycle enzyme, isocitrate dehydrogenase. While it remains unclear whether this observed outcome is attributable to other unseen pleiotropic effects stemming from the *icd* deletion, our data consistently indicates that the most significant reduction in persistence levels occurs with disruptions in energy metabolism. A comprehensive approach, encompassing omics and knockout screening as presented in this study, offers a more complete understanding, revealing the consensus behavior within the entire metabolic network.

Reactive oxygen species (ROS) have been proposed to contribute to antibiotic killing; however, our prior work using identical experimental conditions demonstrated that ROS are unlikely to be a major factor in persister formation during the late stationary phase (Orman & Brynildsen, 2015). In that study, we overexpressed catalytically active antioxidant enzymes, including catalases (KatE, KatG) and superoxide dismutases (SodA, SodB, SodC), yet observed no significant change in persister levels. To further decouple ROS from respiratory activity in that study, we performed anaerobic experiments using nitrate as an alternative terminal electron acceptor. Interestingly, anaerobic respiration enhanced persister formation, and inhibition of nitrate reductases with KCN reduced it—further supporting a ROS-independent mechanism. Together, these findings suggest that under our experimental conditions, it is respiratory activity rather than ROS production that plays a more central role in antibiotic tolerance.

In conclusion, a significant gap in the current literature is the lack of a comprehensive understanding of how bacterial cell metabolism undergoes changes during the transition to a tolerant state. Identifying the specific metabolic pathways that gain significance for cell survival in

this context is crucial. This knowledge can pave the way for the development of more informed and targeted treatment strategies, ultimately enhancing our ability to combat tolerant cells and improve overall treatment outcomes.

Materials and methods

Bacterial Strains and Plasmids

All experiments were conducted using *E. coli* K-12 MG1655 wild-type (WT) and its derivative strains. *E. coli* K-12 MG1655, MO strains (carrying the mCherry expression system), *hipA7* (high persister strain) and pUA66 plasmids were obtained from Mark P. Brynildsen at Princeton University. *E. coli* MO strain was used to monitor cell proliferation at single cell level due to its chromosomally integrated isopropyl β -D-1-thiogalactopyranoside (IPTG)-inducible mCherry expression cassette (Orman & Brynildsen, 2013b [DOI](#), 2013a [DOI](#), 2015 [DOI](#)). *E. coli* K-12 BW25113 WT and single deletions were obtained from Dharmacon Keio Collection (Dharmacon, Catalog# OEC4988, Lafayette, CO, USA). The mutant strains in this study were generated using the Datsenko-Wanner method (Datsenko & Wanner, 2000 [DOI](#)). The pUA66-EV was generated by the removal of *gfp* gene from the plasmid. The *crp* gene with its promoter was cloned into the modified pUA66 plasmid to obtain the pUA66-*crp* expression system. The *sdhABCD* reporter (pMSs201-P_{*sdhABCD*}-*gfp*) and the *cyaA* reporter (pMSs201-P_{*cyaA*}-*gfp*) were obtained from a previous study (Zaslaver et al., 2006 [DOI](#)). The cloning method was followed according to a standard method from NEB (Tirabassi & Bio, 2014 [DOI](#)). Genetic modifications were verified by PCR and gene sequencing (Genewiz, South Plainfield, NJ, USA). A complete list of strains, plasmids, and oligonucleotides used in this study is presented in the **Key Resources Table**.

Media, Chemicals, and Culture Conditions

All chemicals used in this study were purchased from Fisher Scientific (Atlanta, GA, USA), VWR International (Pittsburgh, PA, USA), or Sigma Aldrich (St. Louis, MO, USA). Luria-Bertani (LB) medium was prepared by combining 5 g of yeast extract, 10 g of tryptone, and 10 g of sodium chloride in 1 L of autoclaved deionized (DI) water. LB agar media was prepared by mixing 40 g of pre-mixed LB agar with 1 L of autoclaved DI water; LB agar media were used to enumerate colony-forming units (CFUs) (Amato et al., 2013 [DOI](#); Keren, Kaldalu, et al., 2004 [DOI](#); Orman & Brynildsen, 2015 [DOI](#)). For washing cells and removing chemicals and antibiotics before plating on agar media, 1X Phosphate-Buffered Saline (PBS) was employed. In the persister assay, concentrations of 5 μ g/mL of ofloxacin (OFX), 200 μ g/mL of ampicillin (AMP), and 50 μ g/mL of gentamicin (GEN) were used (Allison et al., 2011 [DOI](#); De Groote et al., 2009 [DOI](#); Keren, Kaldalu, et al., 2004 [DOI](#); Keren, Shah, et al., 2004 [DOI](#)). The retention of plasmids necessitated 50 μ g/mL of kanamycin (KAN) in the culture media (Orman & Brynildsen, 2015 [DOI](#)). Fluorescent protein expression was induced using 1 mM IPTG (Orman & Brynildsen, 2015 [DOI](#)). Overnight pre-cultures were prepared in 14 mL Falcon test tubes containing 2 mL of LB medium, inoculated from a 25% glycerol cell stock stored at -80°C, and incubated for 24 hours at 37°C with shaking at 250 revolutions per minute (rpm). Main cultures were established by diluting the overnight pre-cultures at a ratio of 1:1000 into 2 mL fresh LB medium in 14 mL Falcon test tubes. Experimental cell cultures were prepared by further dilution of the main cultures into either 25 mL fresh LB medium in 250 mL baffled flasks or 2 mL fresh LB medium in 14 mL Falcon test tubes. Cultures at t=5 h and t=24 h were defined as early and late stationary phase cultures, respectively. Detailed experimental procedures are outlined below.

Cell growth and persister assays

Main cultures were prepared by diluting the overnight pre-cultures at a ratio of 1:1000 into 2 mL of fresh LB medium in 14 mL Falcon test tubes. These cultures were then incubated at 37°C with shaking at 250 rpm. Cell growth was monitored by measuring the optical density at 600 nm wavelength (OD₆₀₀) using a Varioskan LUX Multimode Microplate Reader (Thermo Fisher,

Waltham, MA, USA). The plate reader data was collected using Skanlt Software V 5.0. Cell cultures at both early and late stationary phases were collected from the test tubes and transferred to the baffled flasks to achieve $\sim 5 \times 10^7$ cells/mL. This concentration represents an approximately 100-fold dilution of the WT main culture into a fresh medium within the flask. It is important to highlight that we consistently employed flow cytometry to quantify the initial cell count (refer to the section "Monitoring cell division" for comprehensive details). As needed, adjustments in cell number-to-volume were executed to ensure the same cell number among both WT and mutant strains. These cultures were then treated with antibiotics at the indicated concentrations and were cultured with shaking at 37°C for 20 hours. After treatment, a 1 mL sample from each flask was transferred to a microcentrifuge tube for a rigorous washing protocol to minimize antibiotic carryover. The samples were centrifuged at 13,300 RPM ($17,000 \times g$) for 3 minutes, and $>950 \mu\text{L}$ of supernatant was carefully removed without disturbing the pellet. The pellet was then resuspended in $950 \mu\text{L}$ of PBS, achieving a >20 -fold dilution of residual antibiotics. This wash step was repeated once more, resulting in a cumulative >400 -fold dilution. After the final wash, cells were resuspended in $100 \mu\text{L}$ of PBS. A $10 \mu\text{L}$ aliquot of this suspension was serially diluted in a 96-well round-bottom plate, and $10 \mu\text{L}$ from each dilution was spotted onto an agar plate. The remaining $90 \mu\text{L}$ of the sample was also plated to quantify CFU levels around the limit of detection, (~ 1 CFU/mL). Plates were incubated at 37 °C for 16 hours (or up to 48 hours) to allow CFUs to develop. When necessary, this washing procedure was repeated up to six times to assess its impact on CFU recovery.

Mid-exponential phase persister assays

Overnight cultures of *E. coli* K-12 MG1655 or BW25113 WT, and mutant strains were diluted 1:100 in 2 mL of LB medium in 14 mL Falcon tubes and incubated at 37°C with shaking at 250 rpm. At mid-exponential phase ($\text{OD}_{600} \sim 0.25$), cells were challenged with the indicated antibiotics and concentrations. At designated time points, cells (1 mL) were collected, washed with 1X PBS, and plated on agar to determine colony-forming units (CFU). To quantify the initial cell count, $10 \mu\text{L}$ of the culture before treatments was serially diluted and plated on LB agar. CFU levels were assessed after incubating the plates for at least 16 hours at 37°C.

cAMP profile assay

An overnight pre-culture of *E. coli* K-12 MG1655 WT was prepared in test tubes with 2 mL of LB medium. The culture was incubated at 37°C with shaking at 250 rpm for 24 hours. For the experimental cultures, a 1:1000 dilution was made in fresh medium. At time-points 0, 2, 4, 6, 8 and 24 h, $100 \mu\text{L}$ of the experimental culture was washed with cold 1X PBS. After centrifuging at 13,300 rpm ($17,000 \times g$), the supernatant was removed. The cells were then resuspended in $100 \mu\text{L}$ of cell lysis buffer from the Cyclic AMP XP® Assay Kit (Catalog# 4339S, Cell Signaling Technology, Danvers, MA, USA) on ice for 10 minutes. Next, $50 \mu\text{L}$ of lysed cells was mixed with the kit's horseradish peroxidase (HRP)-linked cAMP solution in the cAMP assay plate. This mixture was incubated at room temperature for 3 hours on a plate shaker at 250 rpm. Following the 3-hour incubation, the plate content was discarded, and the plate was washed four times with the kit's Wash Buffer. Then, $100 \mu\text{L}$ of tetramethylbenzidine (TMB) substrate was added to allow color development, and after 30 minutes, $100 \mu\text{L}$ of the stop solution provided by the kit was added. The absorbance was measured at 450 nm using a plate reader. The standard curve was prepared using the same conditions and the standard cAMP solutions provided by the kit to determine cAMP concentrations.

Metabolomics

Metabolites from both Δcrp and WT cells were analyzed at the Metabolon, Inc., facility (Morrisville, NC, USA). Cells were cultured until the early stationary phase and late stationary phase at 37°C with shaking at 250 rpm. Afterward, cells were collected through centrifugation at 4700 rpm at 37°C for 15 minutes, yielding a pellet of approximately $100 \mu\text{L}$ containing around 10^{10}

cells. Subsequently, the cells were washed once with 1X PBS and then centrifuged (13,000 rpm, 3 minutes at 4°C). Following this, they were frozen in an ethanol/dry ice bath for 10 minutes. The extracts from both mutant and WT cells were subjected to analysis using ultra-high-performance liquid chromatography-tandem accurate mass spectrometry (MS), a process aimed at identifying a wide range of metabolites. Sample extraction, preparation, instrument settings, and conditions for the MS platform adhered to Metabolon's protocols (as detailed in our previous study) (Mohiuddin et al., 2022 [DOI](#)). To identify the sample's metabolites among potential false positives from instrument noise, process artifacts, and redundant ion features, the results were cross-referenced with Metabolon's extensive metabolite library (standards). Data normalization was carried out based on protein concentration, determined using the Bradford assay. The significant difference between mutant and WT was identified using Welch's two-sample t-test. Further analysis involved pathway enrichment assessment using MetaboAnalyst (Lu et al., 2023 [DOI](#)). This involved inputting the upregulated and downregulated metabolites based on chosen thresholds. A comprehensive overview of metabolite measurements, pathway enrichment, statistical analyses, and data representations can be found in our previously published study (Mohiuddin et al., 2022 [DOI](#)).

Proteomics

Overnight cultures for both *E. coli* K-12 MG1655 WT and Δcrp strains were prepared using 2 mL of LB medium. Incubation was carried out at 37°C and 250 rpm for 24 hours. The following day, the main cultures were established under the same conditions, using a 1000-fold dilution of the overnight culture in 2 mL fresh LB medium. After 24 hours, the OD₆₀₀ of both WT and mutant strains was measured and adjusted to an OD₆₀₀ of 2.5. For further processing, 2 mL of the main culture was washed twice with cold 1X PBS, maintaining the cold environment throughout. Centrifugation conditions were set at 4°C, 13,000 rpm for 3 minutes. Before the final centrifugation, a cell count was conducted using flow cytometry. This involved using 10 µL of washed culture and 990 µL of 1X PBS. Subsequent to centrifugation, the pellets were collected. Cell lysis was carried out using 300 µL of NEBExpress® *E. coli* Lysis Reagent (Catalog# P8116S, Ipswich, MA, USA) at room temperature for 30 minutes. Following this, the lysed samples were centrifuged at 16,600 x g for 10 minutes, and 250 µL of supernatants were collected from each sample for the assay. The total protein concentration of the supernatants was determined using the bicinchoninic acid (BCA) assay (Catalog# 23225, Thermo Fisher Scientific, Waltham, MA, USA). In a 96-well plate, 25 µL of each cell lysate sample, diluted 5 and 10 times with ultra-pure DI water, were loaded into each well. Subsequently, 200 µL of the BCA working reagent (50:1, Reagent A:B) was added to each well and mixed on a plate shaker for 30 seconds. The plate was then incubated at 37°C for 30 minutes, followed by cooling for 5 minutes at room temperature, shielded from light. The absorbance was finally measured at 562 nm using a plate reader, and the total protein concentration in each sample was calculated using a standard curve prepared from standard protein solutions. Protein analysis for both WT and mutant was conducted by the proteomics service at UT Health's Clinical and Translational Proteomics Service Center (Houston, TX). The samples underwent acetone precipitation, during which proteins were precipitated by exposing them to -20°C for 3 hours. Following this, a centrifugation step at 12,000 x g for 5 minutes separated the precipitated pellets. These pellets were subsequently subjected to denaturation and reduction using a mixture containing 30 µL of 6 M urea, 20 mM DTT in 150 mM Tris HCl (pH 8.0) at 37°C for 40 minutes. Afterward, alkylation was carried out with 40 mM iodoacetamide in the absence of light for 30 minutes. To prepare for digestion, the reaction mixture was diluted 10-fold using 50 mM Tris-HCl (pH 8.0) and then incubated overnight at 37°C with trypsin at a 1:30 enzyme-to-substrate ratio. The digestion process was terminated by adding an equal volume of 2% formic acid, followed by desalting using Waters Oasis HLB 1 mL reverse phase cartridges, following the vendor's recommended procedure. Finally, the elutes were dried using vacuum centrifugation. Approximately 1 µg of the tryptic digest, prepared in a solution containing 2% acetonitrile and 0.1% formic acid in water, underwent analysis using LC/MS/MS. The instrument used was the Orbitrap Fusion™ Tribrid™ mass spectrometer by Thermo Scientific™, connected to a Dionex UltiMate 3000 Binary RSLCnano System. The separation of peptides occurred on an analytical C18 column with dimensions of 100 µm ID x 25 cm, featuring 5 µm particles and an 18 Å pore size.

Peptides were eluted at a flow rate of 350 nL/min. The gradient conditions applied were as follows: a gradient starting from 3% B and increasing to 22% B over a duration of 90 minutes, followed by a step to 22%-35% B for 10 minutes, then another step to 35%-90% B for 10 minutes, and finally, maintaining 90% B for an additional 10 minutes (Solvent A was composed of 0.1% formic acid in water, while solvent B contained 0.1% formic acid in acetonitrile). The peptides were analyzed using a data-dependent acquisition method. The Orbitrap Fusion MS operated by measuring FTMS1 spectra with a resolution of 120,000 FWHM, scanning in the m/z range of 350-1500, using an AGC target set to 2E5, and with a maximum injection time of 50 ms. Within a maximum cycle time of 3 sec, ITMS2 spectra were collected in rapid scan mode. High Collision Dissociation (HCD) was employed with a normalized collision energy (NCE) of 34, an isolation window of 1.6 m/z , an AGC target set to 1E4, and a maximum injection time of 35 ms. Dynamic exclusion was implemented for a duration of 35 sec to prevent repeated analysis of the same ions. For the experimental analysis, the Thermo Scientific™ Proteome Discoverer™ software version 1.4 was utilized to process the raw data files. The spectra were subjected to analysis against the *E. coli* proteome database (Swiss-Prot 29,161) through the Sequest HT search engine. Additionally, the spectra were compared against a decoy database, employing a target false discovery rate (FDR) of 1% for stringent criteria and 5% for more relaxed criteria. The enzymatic cleavage allowance for trypsin included up to two potential missed cleavages. The MS tolerance was defined as 10 ppm, while the MS/MS tolerance was set at 0.6 Da. Fixed modification involved carbamidomethylation on cysteine residues, and variable modifications encompassed methionine oxidation and asparagine deamidation. For proteomics data processing and fold change calculations, the approaches were essentially followed by the method paper from Aguilan et al. (Aguilan et al., 2020 [DOI](#)). Then, the STRING tool V12.0 (Szklarczyk et al., 2021 [DOI](#), 2023 [DOI](#)) was employed to find the significant networks among input proteins. To generate the protein network and pathway enrichment analysis, we input the protein identifiers (Accession numbers) for upregulated or downregulated proteins with at least a 2-fold increase or reduction, respectively. *E. coli* K-12 was selected as the organism of interest. We opted for evidence as the criterion for network edges, prioritizing the type of interaction evidence, helping us conduct an automated pathway-enrichment analysis, centering on the entered proteins and identifying pathways that occurred more frequently than expected. This analysis was grounded in the statistical background of the entire genome and encompasses various functional pathway classification frameworks, such as Gene Ontology annotations, KEGG pathways, and Uniprot keywords, as detailed elsewhere (Szklarczyk et al., 2021 [DOI](#), 2023 [DOI](#)). In pathway enrichment analysis, the "strength score," calculated as $\text{Log}_{10}(\text{observed/expected})$, serves to assess the degree or significance of enrichment within a specific biological pathway. This metric reflects the magnitude of the enrichment effect, with a higher score indicating stronger enrichment. The score is derived from the ratio of i) annotated proteins in the network for a given term to ii) the expected number of proteins annotated with the same term in a random network of equivalent size (Szklarczyk et al., 2021 [DOI](#), 2023 [DOI](#)). To gauge the significance of enrichment, False Discovery Rate (FDR) is employed. FDR scores represent p-values corrected for multiple testing within each category using the Benjamini-Hochberg procedure (Szklarczyk et al., 2021 [DOI](#), 2023 [DOI](#)).

Monitoring cell division

To monitor cell division and quantify non-growing cells, we utilized inducible fluorescent protein (mCherry) expression. Overnight pre-cultures of *E. coli* MO were prepared with 2 mL of LB medium containing 1 mM of IPTG. These cultures were grown in test tubes at 37°C with shaking at 250 rpm for 24 hours. Main cultures were established by diluting the overnight pre-cultures (at a ratio of 1:1000) into 2 mL of fresh LB medium in 14 mL Falcon test tubes. These cultures were incubated at 37°C with shaking at 250 rpm. Cells were allowed to grow until they reached the early stationary phase and the late stationary phase. The mCherry positive cells were then collected, washed twice with 1X PBS to remove the IPTG from the culture, and subsequently re-suspended in fresh 2 mL LB media in test tubes to achieve $\sim 5 \times 10^7$ cells/mL. This concentration represents an approximately 100-fold dilution. When needed, adjustments in cell number-to-volume were made to ensure the same cell number among both WT and mutant strains. In the experimental culture

test tubes, 2 mL of LB medium was added, and the volume of washed cells was inoculated to achieve an OD₆₀₀ of 0.0286. The culture was then incubated at 37°C with shaking at 250 rpm. At specific time points (0, 1, 2, and 2.5 h), cells were collected and re-suspended in 1X PBS to measure their fluorescent protein content using flow cytometry. For flow cytometry analysis, cells were collected and diluted to a desired cell density ($\sim 10^6$ - 10^7 cells/mL) in 1 mL of 1X PBS in flow cytometry tubes (5 mL round-bottom Falcon tubes, size: 12 x 75 mm). The flow cytometry analysis was conducted using a NovoCyte 3000RYB instrument (ACEA Bioscience Inc., San Diego, CA, USA). During flow cytometry analysis, a slow sample flow rate of 14 μ L/min was chosen, along with a sample stream diameter (core diameter) of 7.7 μ m. The instrument maintained a constant sheath flow rate of 6.5 mL/min. The core diameter was calculated using the ratio of the sample flow rate to the sheath flow rate. These specific conditions were selected to achieve improved data resolution for the size of *E. coli* cells. Flow diagrams utilized forward and side scatter signals from viable cells, alongside a control of solvent devoid of cells, to ascertain the presence of cells. For the flow cytometry analysis, cells were excited at a 561 nm wavelength, and the red fluorescence was detected using a 615/20 nm bandpass filter.

Flow Cytometry Analysis of Cell Growth

Overnight cultures of *E. coli* K-12 MG1655 MO WT and mutant strains were diluted at a ratio of 1:1000 into 2 mL of fresh LB medium, placed in 14 mL Falcon test tubes, and incubated at 37°C with shaking at 250 rpm for 24 hours. For the main cultures, a similar strategy was employed. The cultures were diluted at a ratio of 1:100 into 2 mL of fresh LB medium in 14 mL Falcon test tubes. These cultures were then incubated at 37°C with shaking at 250 rpm. At specific time points, including t=0, 20 min, 40 min, and 1 to 5 h, the cell growth was halted. This was achieved by diluting the cells in 1X PBS containing 25 μ g/mL of chloramphenicol (CAM). The CAM treatment allowed for subsequent analysis without further division. Flow cytometry was then utilized to measure the number of cells at each of these time points. This approach provided insight into cell division dynamics and allowed for the quantification of cell populations under specific conditions.

Fluorescent protein expression assay for reporter genes

Mutant and control strains were derived from *E. coli* K-12 MG1655 and carried pMSs201-*gfp* plasmids incorporating P_{sdhABCD} or P_{cyaA} gene promoters. Overnight pre-cultures were prepared using 2 mL of LB medium supplemented with 50 μ g/mL KAN. These cultures were incubated in test tubes within a shaker at 37°C for 24 hours. Main cultures were established by diluting the overnight pre-cultures at a ratio of 1:1000 into 2 mL of fresh LB medium within 14 mL Falcon test tubes. These main cultures were maintained at 37°C with shaking at 250 rpm. Cell cultures at the desired growth phase were collected and then diluted to attain a desired cell density of around 10^6 - 10^7 cells/mL in 1 mL of 1X PBS within flow cytometry tubes. This allowed for subsequent flow cytometry analysis, using the same conditions as described earlier for monitoring cell division (refer to "Monitoring cell division"). During analysis, a laser emitting light at 488 nm was used to excite the cells, and the resulting green fluorescence was detected using a 530/30 nm bandpass filter. This setup enabled the examination of the fluorescence patterns of the cells, offering insights into their dynamics under different conditions.

Redox Sensor Green assay

To gauge bacterial metabolic activity, we employed the Redox Sensor Green (RSG) dye from Thermo Fisher (Catalog# B34954, Thermo Fisher Scientific, Waltham, MA). *E. coli* K-12 MG1655 WT and mutant cells from the desired growth phase were diluted at a ratio of 1:100 in 1 mL of 1X PBS. To this solution, 1 μ L of the RSG dye was added to flow cytometry tubes. After a brief vortexing to ensure uniform mixing, the samples were incubated at 37°C in darkness for 10 minutes. Subsequently, these samples were subjected to flow cytometry analysis. For the flow cytometry analysis, the same methodology as employed in "Monitoring cell division" was followed, with one variation. Cells were excited at 488 nm during analysis, and the resulting green fluorescence was

detected using a 530/30 nm bandpass filter. This setup allowed us to assess the fluorescence patterns, reflecting the metabolic activity of the bacterial cells under different conditions. As a control measure, cells were treated with 20 μ M of carbonyl cyanide m-chlorophenyl hydrazone (CCCP) for 5 minutes before the addition of the RSG dye. This served to validate the assay's sensitivity to changes in metabolic activity (**Figure 5—figure supplement 1** [↗](#)).

Metabolic activity of non-lysing cells and VBNC cell quantification

Overnight cultures of *E. coli* K-12 MG1655 MO WT and mutant strains were diluted at a ratio of 1:1000 into 2 mL of fresh LB medium supplemented with 1 mM IPTG. These cultures were established in 14 mL Falcon test tubes and incubated at 37°C with shaking at 250 rpm for 24 hours. Treatment cultures were prepared by diluting the main cultures at a ratio of 1:100 into 25 mL of fresh LB medium supplemented with 1 mM IPTG. These cultures were set up in 250 mL baffled flasks and contained 200 μ g/mL AMP. They were then cultured at 37°C with shaking at 250 rpm for 20 hours. Both before and after the treatment, 1 mL samples were collected from the cultures. These samples were subjected to a washing procedure with 1X PBS to eliminate the antibiotic present in the samples. The washed cells were then resuspended in 1 mL of 1X PBS within flow cytometry tubes. To measure the metabolic activity of the non-lysing cells, the RSG dye was employed as described above. Intact cells (following antibiotic treatment), stained as live with RSG, comprised both persister and VBNC cells. Persister levels were quantified by plating the cells on an agar medium, as described previously ([Orman & Brynildsen, 2013b](#) [↗](#)). As VBNC cells cannot grow on agar medium, their enumeration involved subtracting the number of persister cells from the total number of intact cells.

Screening *E. coli* (K-12 BW25113) Keio Knockout Collection

Overnight cultures of individual mutant strains, along with their parental strain K-12 BW25113 WT harboring a kanamycin-resistant marker, were diluted at a ratio of 1:1000 in fresh LB medium containing 50 μ g/mL of KAN. This was done in 14 mL Falcon test tubes and the cultures were then incubated at 37°C with shaking at 250 rpm. Upon reaching the late stationary phase, cells were further diluted at a ratio of 1:100 in fresh medium supplemented with antibiotics at specified concentrations. These cultures were once again incubated at 37°C with shaking for 20 hours. Following the 20-hour treatment period, the same methodology described in the section "Cell growth and persister assays" was employed to quantify the number of persisters. This approach allowed for an assessment of the impact of antibiotics on the formation of persister cells for both mutant strains and the parental K-12 BW25113 WT strain.

Persister quantitation in *E. coli* K-12 MG1655 single gene deletions

Overnight cultures of mutant strains were diluted at a ratio of 1:1000 in 14 mL Falcon test tubes containing 2 mL of LB medium. These cultures were then incubated at 37°C with shaking at 250 rpm. Upon reaching the late stationary phase, cells were diluted at a ratio of 1:100 in fresh medium supplemented with antibiotics at specified concentrations. The cultures were once again subjected to shaking at 37°C for 20 hours. The same method described earlier, referred to as "Cell growth and persister assays," was employed to quantify the number of persister cells resulting from this treatment. This approach allowed for the assessment of the impact of antibiotic exposure on persister cell formation within the mutant strains.

Statistics and reproducibility

A nonlinear logarithmic model was employed to create biphasic kill curves ([Mohiuddin, Hoang, et al., 2020](#) [↗](#); [Windels et al., 2019](#) [↗](#)). The significance of these kill curves was determined through the utilization of F-statistics ([Mohiuddin, Hoang, et al., 2020](#) [↗](#); [Windels et al., 2019](#) [↗](#)).

Metabolomics data were subjected to analysis using Welch's two-sample t-test in order to identify metabolites that significantly differed between the control and mutant groups ([YUEN, 1974](#) [↗](#)). For all experiments, a minimum of three independent biological replicates were conducted, unless

explicitly stated otherwise. In each figure (excluding flow diagrams), the data for each time point are represented as the mean value accompanied by the standard deviation. In terms of statistical significance analysis, the designated threshold values for P were set as follows: *P < 0.05, **P < 0.01, ***P < 0.001, and ****P < 0.0001. All figures were generated using GraphPad Prism 10.3.0. The statistical analyses were carried out using the statistical functions of GraphPad Prism 10.3.0. For the clustering of metabolomics and proteomics data, the "Clustergram" function of MATLAB (V R2020b) was employed. FlowJo (V 10.8.1) was the tool used to analyze the data acquired from flow cytometry.

Data Availability

All data generated or analyzed during this study are included in the manuscript and supporting files.

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Supplementary Figures

Figure 1—figure supplement 1.

cAMP concentrations normalized to cell numbers for *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$.

The cAMP levels were measured in late stationary phase cultures at 450 nm using the Cyclic AMP XP® Assay Kit (Cell Signaling Technology). $n=4$. Statistical significance was observed between control and mutant strains (* $P < 0.05$, *** $P < 0.001$, One-way ANOVA with Dunnett's multiple comparisons test). The data for each time point represent the mean value $\pm$ standard deviation.

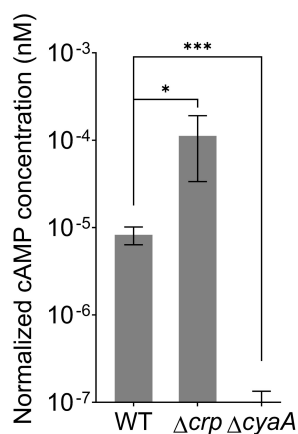


Figure 1—figure supplement 2.

Genetic perturbation of Crp/cAMP enhanced P_{cyaA} promoter activity, resulting in increased *gfp* expression.

Overnight cultures of *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$ strains harboring the pMSs201 plasmid, which encodes green fluorescent protein (GFP) under the control of the P_{cyaA} promoter, were diluted 1:1000 into fresh LB medium and incubated at 37°C with shaking at 250 rpm for 24 hours. Cells at the late stationary phase were then collected, diluted in 1X PBS, and analyzed by flow cytometry. $n=4$. Statistical significance was observed between control and mutant strains (**** $P < 0.0001$, One-way ANOVA with Dunnett's multiple comparisons test). The data for each time point represent the mean value $\pm$ standard deviation.

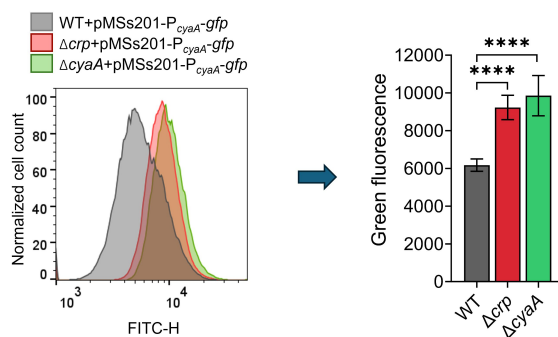


Figure 1—figure supplement 3.

Growth curves of *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$.

Optical densities of cell cultures at 600 nm (OD_{600}) were measured every hour using a plate reader. $n=3$. The data for each time point represent the mean value $\pm$ standard deviation.

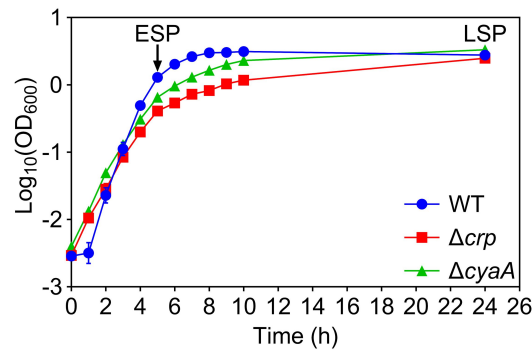
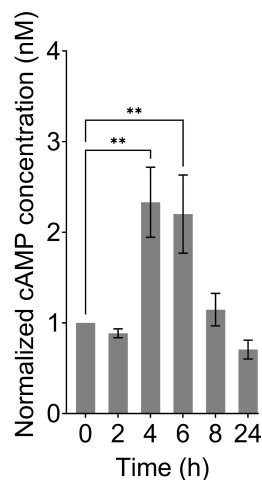


Figure 1—figure supplement 4.

Normalized cAMP concentrations of *E. coli* K-12 MG1655 WT.

The cAMP concentrations were measured in growth cultures at the indicated time points. First, the cAMP concentrations were normalized to the number of cells. Subsequently, the data were further normalized based on the time point 0 to mitigate errors associated with batch-to-batch assay kit variations and to capture the trend in cAMP levels across the time points. $n=8$. Statistical significance was observed between time points (** $P < 0.01$, One-way ANOVA with Dunnett's multiple comparisons test). The data for each time point represent the mean value $\pm$ standard error.



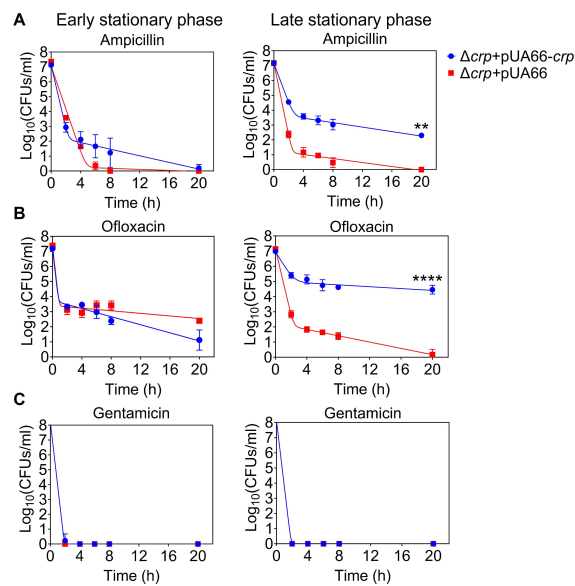


Figure 1—figure supplement 5.

Persister levels of cells carrying the Crp expression system.

Cells at early (t=5 h) and late (t=24 h) stationary phases were transferred to fresh media with antibiotics for persister cell quantification. At time points 0, 2, 6, 8, and 20 h, 1 mL of the treated culture underwent two washes with 1X PBS to remove antibiotics. It was then serially diluted and plated on an agar plate to count the CFUs. **(A)** Persister levels of ampicillin-treated culture (200 µg/mL). **(B)** Persister levels of ofloxacin-treated cultures (5 µg/mL). **(C)** Persister levels of gentamicin-treated culture (50 µg/mL). n=4. Biphasic kill curves were generated using a non-linear model (see **Materials and Methods**). Statistical significance tests were conducted using F-statistics (**P < 0.01 and ****P < 0.0001). The data for each time point represent the mean value ± standard deviation.

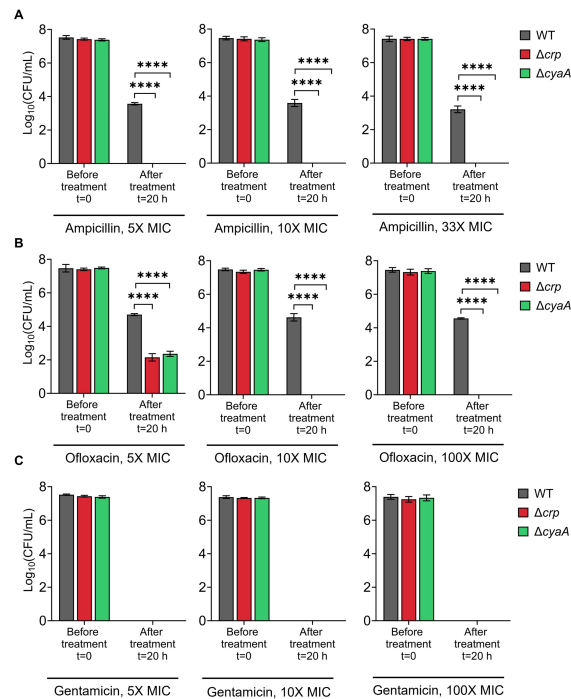


Figure 1—figure supplement 6.

Persister levels of *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyoA$ strains at normalized antibiotic concentrations.

Panels show survival following treatment with (A) ampicillin, (B) ofloxacin, and (C) gentamicin. After treatment, samples were washed six times with 1X PBS to minimize antibiotic carryover. Antibiotic concentrations were normalized to MICs to ensure valid comparisons across strains (The concentrations of 33× MIC for ampicillin and 100× MIC for ofloxacin and gentamicin match those used in [Figure 1](#)). n=4. Statistical significance was observed between control and mutant strains (****P < 0.0001, One-way ANOVA with Dunnett's multiple comparisons test). The data for each time point represent the mean value ± standard deviation.

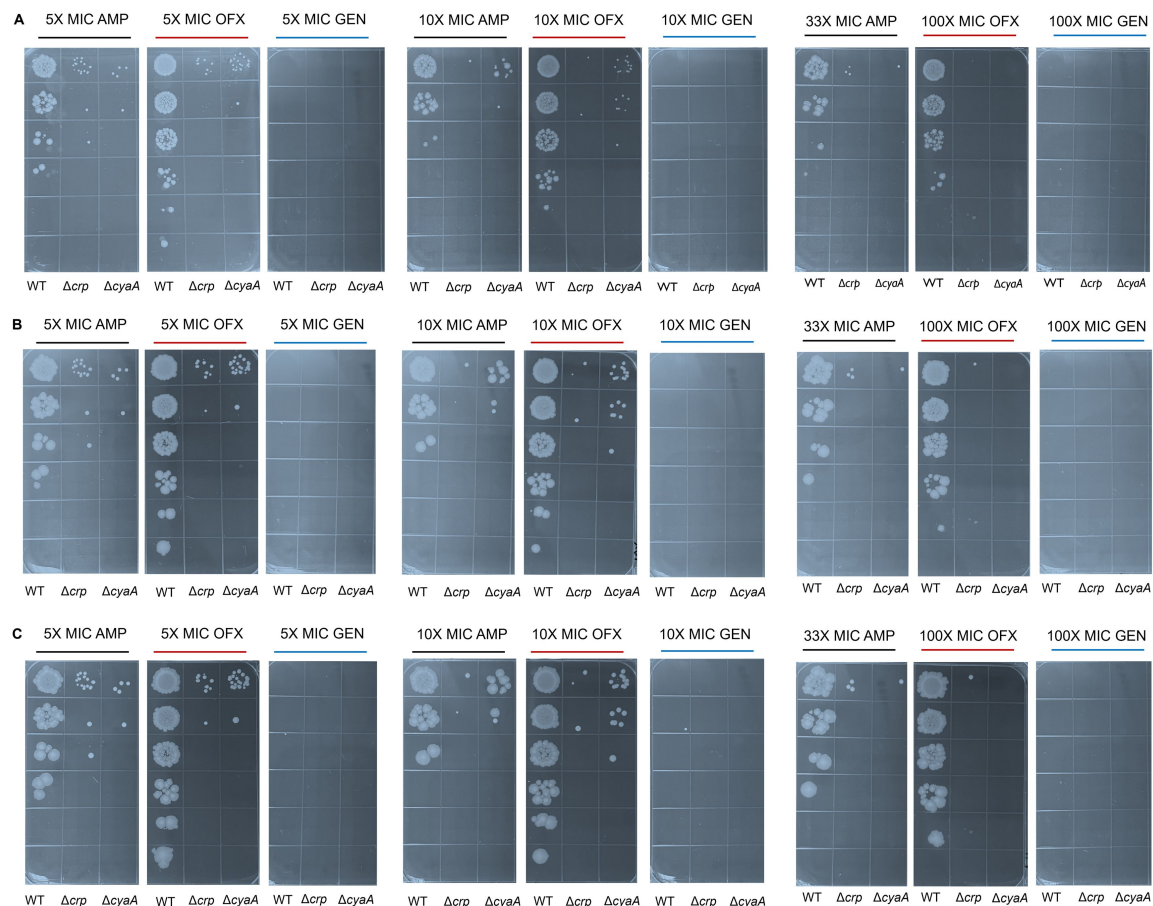


Figure 1—figure supplement 7.

Agar plates showing *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$ strains following treatment with antibiotics at normalized concentrations.

After treatment, cells were washed and subjected to 10-fold serial dilutions, then plated on agar. Plates were incubated for (A) 16 h, (B) 48 h, and (C) 72 h to assess colony formation over time. The panel is a representative biological replicate. Consistent results were seen across all 3 biological replicates.

Figure 1—figure supplement 8.

Deletion of *crp* and *cyaA* reduces ampicillin and ofloxacin persistence in the *hipA7* strain.

Persistence levels were assessed by exposing *E. coli* K-12 MG1655 WT, *hipA7*, *hipA7Δcrp*, and *hipA7ΔcyaA* strains in the late stationary phase to the specified antibiotics, followed by CFU quantification at designated time points. (A) Ampicillin (200 μg/ml), (B) Ofloxacin (5 μg/ml) and (C) Gentamicin (50 μg/ml). n=4. Statistical significance was observed between control and mutant strains (****P < 0.0001, One-way ANOVA with Dunnett's multiple comparisons test). The data for each time point represent the mean value ± standard deviation.

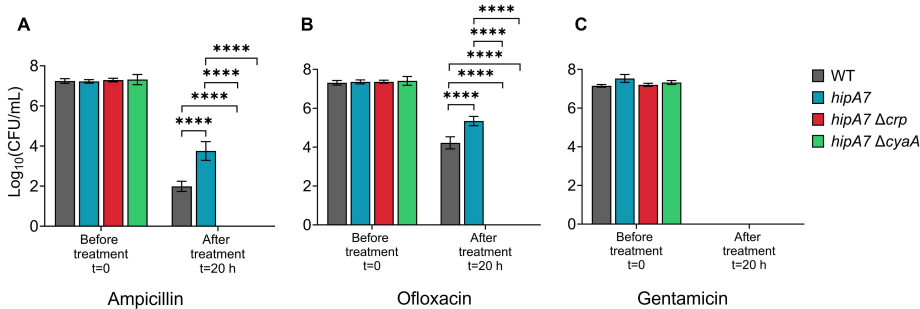


Figure 2—figure supplement 1.

The pathway enrichment analysis comparing WT and *Δcrp* strains during the early stationary phase cultures.

The analysis was conducted using MetaboAnalyst, with a threshold ratio (*Δcrp*/WT) set at ≤ 0.5 for downregulation and ≥ 2 for upregulation. (A) Upregulated and (B) Downregulated pathways in the *Δcrp* strain compared to WT during the early stationary growth phase (ESP).

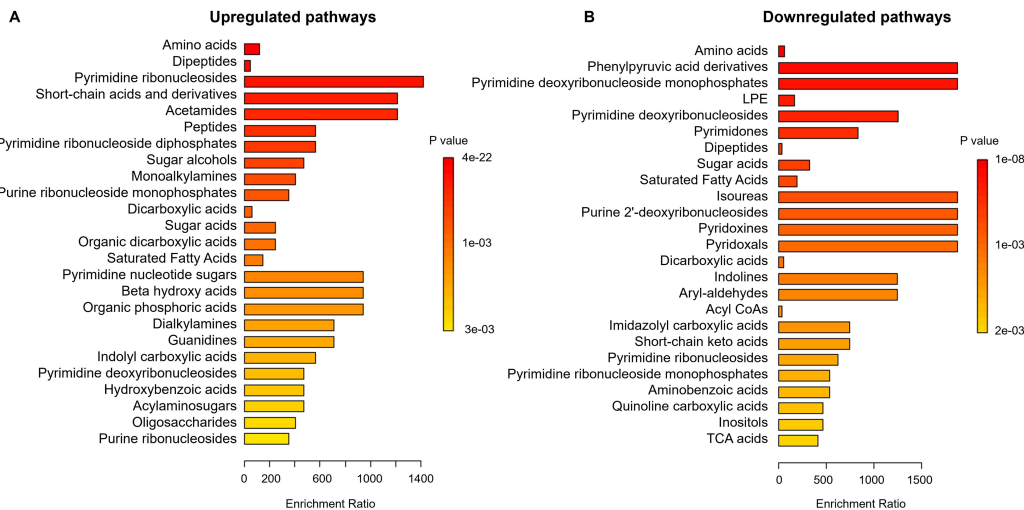


Figure 2—figure supplement 2.

The pathway enrichment analysis for the WT strain.

The analysis was conducted using MetaboAnalyst, with a threshold ratio (LSP/ESP) set at ≤ 0.5 for downregulation and ≥ 2 for upregulation. (A) Upregulated and (B) Downregulated pathways of the WT strain in the late stationary growth phase (LSP) compared to the WT strain in the early stationary phase (ESP).

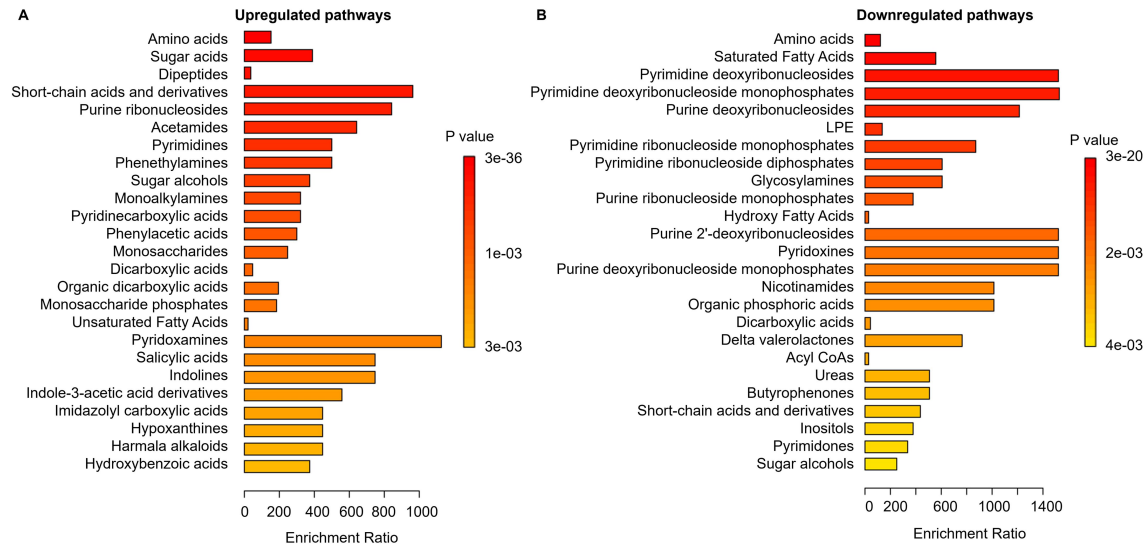


Figure 2—figure supplement 3.

The pathway enrichment analysis for the Δcrp strain.

The analysis was conducted using MetaboAnalyst, with a threshold ratio (LSP/ESP) set at ≤ 0.5 for downregulation and ≥ 2 for upregulation. (A) Upregulated and (B) Downregulated pathways of the Δcrp strain in the late stationary growth phase (LSP) compared to the Δcrp strain in the early stationary phase (ESP).

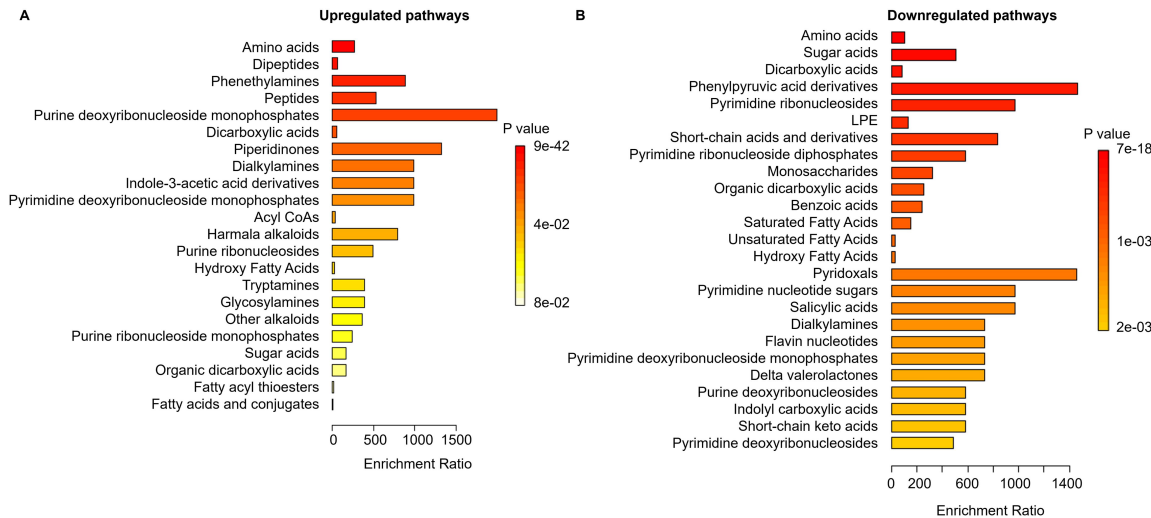


Figure 3—figure supplement 1.

The MS analysis of proteins from both WT and Δcrp strains at the late stationary phase.

The proteomic data were subjected to unsupervised hierarchical clustering. Each column in the figure represents a biological replica. n=3.

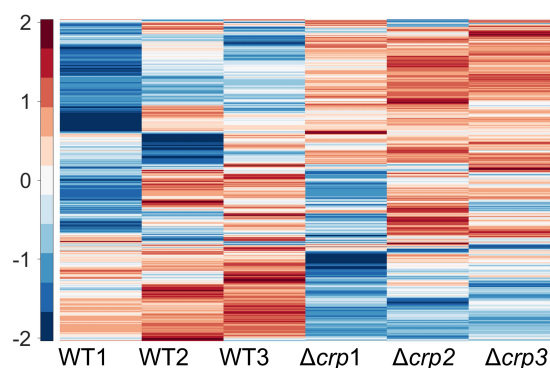
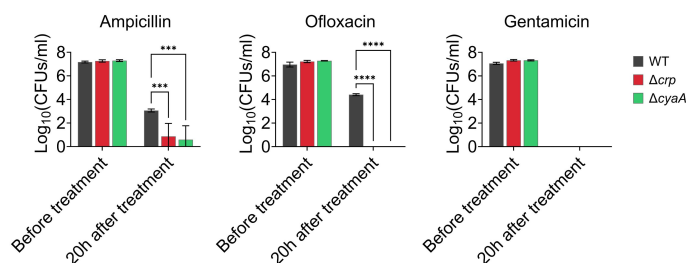


Figure 4—figure supplement 1.

Persister levels of *E. coli* WT, Δcrp , and $\Delta cyaA$ cells with the integrated mCherry expression system.

Late stationary phase cultures (t=24 h) were transferred to fresh media and treated with ampicillin (200 μ g/mL), ofloxacin (5 μ g/mL), and gentamicin (50 μ g/mL) for 20 hours. Subsequently, 1 mL of the treated culture underwent two washes with 1X PBS to remove antibiotics. It was then serially diluted and plated on an agar plate to count the CFUs. The levels of ofloxacin and gentamicin persisters in the mutant strains were below the limit of detection. n=4. Statistical significance was observed between control and mutant strains (***P < 0.001, ****P < 0.0001, Two-way ANOVA with Tukey's multiple comparisons test). The data for each time point represent the mean value $\pm$ standard deviation.



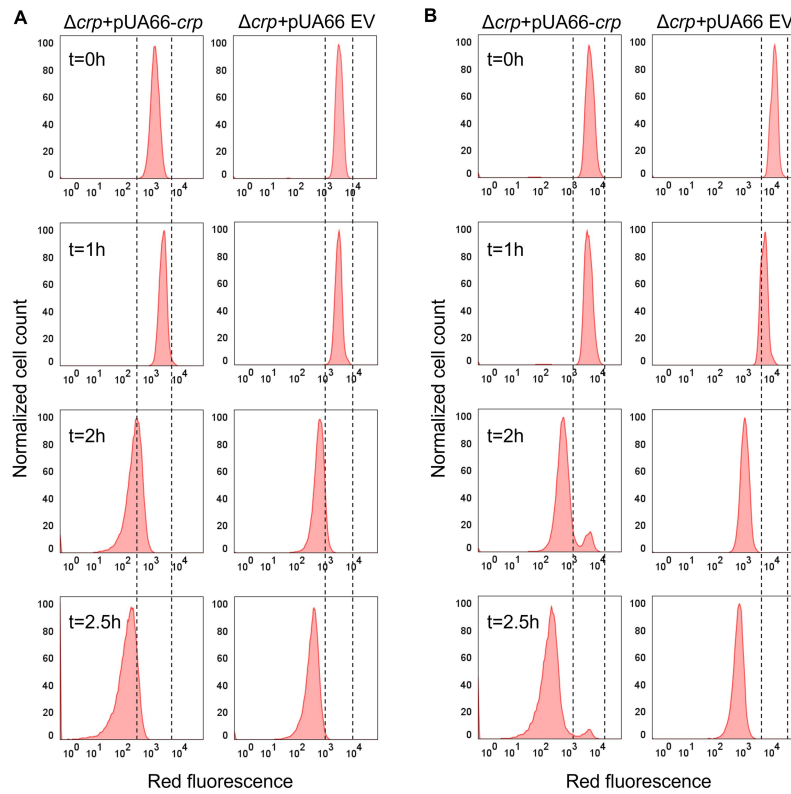


Figure 4—figure supplement 2.

Non-growing cell levels in the *E. coli* strain carrying the Crp expression system.

(A, B) Flow cytometry histograms depict mCherry expression in $\Delta crp+pUA66-crp$, and $\Delta crp+pUA66 EV$ at early ($t=5 h$) and late ($t=24 h$) stationary phases, respectively. Cells containing an IPTG-inducible mCherry expression system were cultivated with IPTG. After washing and dilution of early and late stationary phase cells in IPTG-free fresh media, fluorescence was tracked in non-growing and growing cells for 2.5 hours. The panel is a representative biological replicate. Consistent results were seen across all 3 biological replicates.

Figure 5—figure supplement 1.

RSG staining control for bacterial metabolic activities.

Exponential phase ($t=3$ h) cells were stained with $1\ \mu\text{M}$ RSG for 10 minutes at 37°C before analyzing by flow cytometry. Unstained cells and cells treated with $20\ \mu\text{M}$ CCCP + $1\ \mu\text{M}$ RSG were used as control. CCCP was expected to reduce cellular redox activities. The panel is a representative biological replicate. Consistent results were seen across all 3 biological replicates.

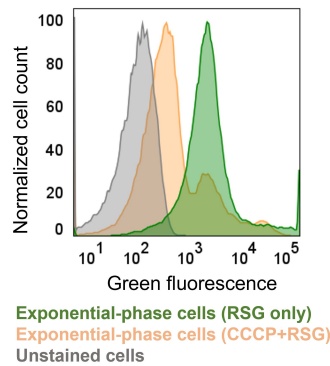


Figure 5—figure supplement 2.

Intact (non-lysed) cell levels of *E. coli* WT, Δcrp , and ΔcyaA cells with the integrated mCherry expression system.

mCherry positive cells were diluted into fresh media and treated with ampicillin ($200\ \mu\text{g}/\text{mL}$) for 20 hours. Flow cytometry was used to measure the red fluorescence of intact surviving cells at time points $t=0$, 1, 2, 4, 6 and 20 h. A representative biological replicate is shown, with consistent results across all 3 replicates.

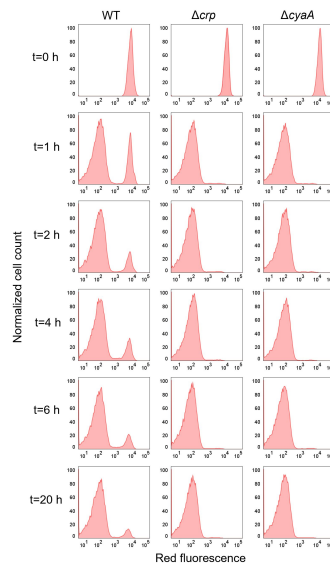


Figure 5—figure supplement 3.

VBNC levels of *E. coli* WT, Δcrp , and $\Delta cyaA$ cells with the integrated mCherry expression system.

Flow cytometry was employed to quantify intact surviving cells. Persister cells were quantified by plating the cells on agar media. Viable but nonculturable (VBNC) cells were enumerated by subtracting persister levels from the intact cell levels. $n=4$. Statistical significance was observed between control and mutant strains (**** $P < 0.0001$, One-way ANOVA using Dunnett's multiple comparisons test). The data for each time point represent the mean value $\pm$ standard deviation.

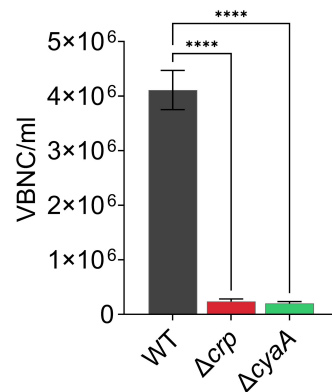
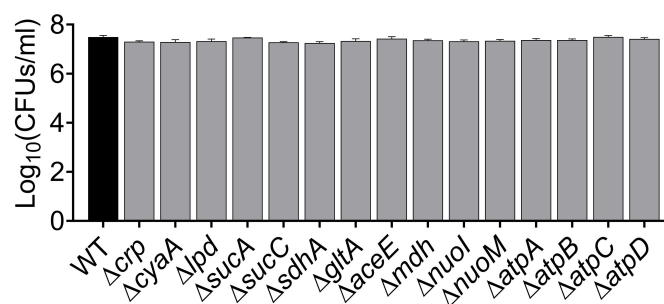


Figure 5—figure supplement 4.

Cell counts of *E. coli* K-12 MG1655 WT and mutant strains using flow cytometry at late stationary phase.

Cells were diluted 100-fold into 1 mL of 1X PBS. $n=4$. The data for each time point represent the mean value $\pm$ standard deviation.



Additional files

Figure 2 - Source Data 1. Normalized Metabolomics Data of Early and Late Stationary Phases of Wild-Type and Mutant *crp* Across 3 Biological Replicates. Data Collected and Analyzed by Metabolon Inc. (Morrisville, NC). [↗](#)

Figure 3 - Source Data 1. Normalized Proteomics Data of the Late Stationary Phase of Wild-Type and Mutant *crp* Across 3 Biological Replicates. Data Collected by UT Health's Clinical and Translational Proteomics Service Center (Houston, TX). [↗](#)

Key Resources Table. [↗](#)

Supplementary File 1. MIC of antibiotics and concentrations of bactericidal antibiotics used in persister assays. [↗](#)

Supplementary File 2. Analyzed metabolomics data. [↗](#)

Supplementary File 3. Analyzed proteomics data. [↗](#)

Supplementary File 4. Supplementary File 4. Analyzed persister survival fraction data. [↗](#)

Supplementary File 5. The knockout strains generated using the *E. coli* K-12 MG1655 background in this study. [↗](#)

Supplementary File 6. Persister levels of *E. coli* K-12 MG1655 WT, Δcrp , and $\Delta cyaA$ strains in late stationary phase. [↗](#)

Supplementary File 7. Persister levels of *E. coli* K-12 MG1655 (A) and BW25113 (B) WT, Δcrp , and $\Delta cyaA$ strains in the exponential growth phase. [↗](#)

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Reviewer #1 (Public review):

The authors set out to understand the role played by a key global metabolic regulator called Crp/cAMP in the formation of persister *Escherichia coli* that survive antibiotic treatment without acquiring genetic mutations.

In order to achieve this aim, the authors employ an interdisciplinary approach integrating standard microbiology assays with cutting-edge genomic, metabolomic and proteomics screening.

The data presented by the authors convincingly demonstrate that the deletion of two key genes that are part of the Crp/cAMP complex (i.e. *crp* and *cyaA*) leads to a significant decrease in the number of *E. coli*.

The authors have carried out additional experiments to further validate this point by using the well characterised *hipA7 E. coli* mutant.

The data presented also demonstrate that deletion of the *crp* gene leads to an overall decrease in energy metabolism and an overall increase in anabolic metabolism at the population level. The deletion of *cyaA* has an opposite effect on cAMP concentration compared to *crp* deletion, the authors presented a possible hypotheses but did not test it.

The authors have now explicitly acknowledged in their discussion that the data presented in this study are obtained at the whole population level rather than at the level of the persister subpopulation and therefore should be considered with caution.

Finally, the authors convincingly show that the persisters they investigated are non-growing and have a higher redox activity and that the deletion of key genes involved in energy metabolism leads to a decrease in the number of persisters.

These data will be important for future investigations on the biochemical mechanisms that allow bacteria to adapt to stressors such as nutrient depletion or exposure to antibiotics. As such this work will likely have an impact in a variety of fields such as bacterial biochemistry, antimicrobial resistance research and environmental microbiology.

Strengths:

Interdisciplinary approach.

Excellent use of replication and ensuring reproducibility.

Excellent understanding and presentation of the biochemical mechanisms underpinning bacterial physiology via an integrated genomic, metabolomic and proteomic screening.

Weaknesses:

There is no tested mechanisms explaining why the deletion of *cyaA* has an opposite effect on cAMP concentration compared to *crp* deletion.

Metabolomics, proteomics and metabolic activity data are obtained at the whole population level rather than at the level of the persister sub-population.

<https://doi.org/10.7554/eLife.99735.2.sa1>

Author response:

The following is the authors' response to the original reviews.

Reviewer #1:

(1) Two genes from the Crp/cAMP complex (crp and cyaA) are hypothesized to be key for persistence but key metabolomics and proteomics data are obtained from only one deletion mutant in the crp gene.

We thank the reviewer for their thoughtful assessment of our manuscript and for providing valuable comments.

In our study, we have demonstrated that deletion of both *cyaA* and *crp* genes results in the same persistence phenotype. In a previous study, we screened knockout strains of global transcriptional regulators using the aminoglycoside (AG) potentiation assay and found that, across a panel of carbon sources, AG potentiation occurred in tolerant cells derived from most knockout strains—except for Δcrp and Δcrp (Mok et al., 2015). This indicated that both genes are critical components of the Crp/cAMP regulatory network in persistence. Because cAMP exerts its effects when bound to its receptor protein Crp, disrupting *crp* alone should effectively abolish Crp/cAMP complex function (Keseler et al., 2011). Thus, we reasoned that comparing Δcrp to wild-type would be sufficient to capture the key metabolic and proteomic alterations arising from Crp/cAMP perturbation. Given the substantial cost and labor intensity of untargeted metabolomics and proteomics analyses, this experimental design

allowed us to extract meaningful insights while maintaining feasibility. Nonetheless, to ensure the robustness of our findings, we have conducted all subsequent validation experiments using both Δcrp and Δcrp strains, confirming that the observed metabolic and proteomic changes are consistent across both mutants. We have now provided a concise justification statement in the manuscript (see lines 197-200 in the current manuscript).

(2) *The deletion of crp and crp have opposite effects on the concentration of cAMP, a comparison of metabolomics and proteomics data obtained using both mutants might aid in understanding this difference.*

Although this is an interesting outcome, we have already discussed in the manuscript that it is likely due to the feedback regulation of the Crp/cAMP complex on crp expression (see Fig. 1 Keseler et al., 2011) (Aiba, 1985; Keseler et al., 2011; Majerfeld et al., 1981). Specifically, perturbation of the Crp/cAMP complex by deleting crp should enhance crp promoter (P_{crp}) activity, leading to increased CyaA protein expression and, consequently, elevated intracellular cAMP levels. To experimentally verify this predicted feedback regulation, we utilized *E. coli* K-12 MG1655 WT, Δcrp , and Δcrp strains harboring the pMSs201 plasmid, which encodes green fluorescent protein (gfp) under the control of the P_{cyaA} promoter. This design allowed us to directly assess the effect of Crp/cAMP perturbation on P_{cyaA} activity by quantifying gfp expression as a reporter. By comparing the mutant strains to WT, we could determine whether loss of Crp/cAMP function indeed derepresses crp expression. As expected, genetic perturbation of Crp/cAMP enhanced P_{cyaA} promoter activity, resulting in increased gfp expression (Figure 1-figure supplement 2). This result supports the role of Crp/cAMP in regulating crp expression via feedback control. We have now explicitly discussed this rationale in the manuscript and included the corresponding data (see lines 410-418 and Figure 1-figure supplement 2 in the current manuscript).

(3) *Metabolomics, proteomics, and metabolic activity data are obtained at the whole population level rather than at the level of the persister sub-population.*

Performing metabolomic, proteomic, and other assays at the level of the persister subpopulation is inherently challenging in this study and across the persister research field, as it requires isolating a pure persister population. While metabolic inhibitors like rifampin and tetracycline can induce dormancy and antibiotic tolerance in the entire population (Kwan et al., 2013), these treatments generate artificially altered cell states that may not accurately reflect naturally occurring persisters. Fluorescent reporters combined with fluorescence-activated cell sorting (FACS) have been utilized to study persister cells, including in our previous studies (Amato et al., 2013; Orman & Brynildsen, 2013, 2015). However, this approach only enriches for persisters rather than isolating a pure population, as persisters still constitute a small fraction of the sorted cells (Amato et al., 2013; Orman & Brynildsen, 2013, 2015). Despite these limitations, our untargeted metabolomics and proteomics analyses at the whole-population level provide valuable insights into the regulatory mechanisms of the Crp/cAMP complex and its potential role in persister formation. We have rigorously examined the impact of these mechanisms on non-growing cell formation (see Figure 4 in the current manuscript) and persister levels (see Figure 5 in the current manuscript) through flow cytometry and single-gene deletion experiments. We appreciate the reviewer's comment and have acknowledged and discussed these methodological challenges in our manuscript (see lines 397-406 in the current manuscript).

Reviewer #2:

(1) *The approaches used here are aimed at the major bacterial population, but yet the authors used the data reflecting the major population behavior to interpret the physiology of persister cells that comprise less than 1% of the major bacterial*

population. How they can pick up a needle from the hay without being fooled by the spill-over artifacts from the major population? Although it is probably very difficult to isolate and directly assay persister cells, firm conclusions for the type proposed by the authors cannot be firmly established without such assays. Perhaps introducing crp/crp mutation into the best example of persistence, the hipA-7 high persistence phenotype may clarify this issue to a certain extent.

We thank the reviewer for their thoughtful assessment of our manuscript and for providing valuable comments.

Performing metabolomics and proteomics at the level of the persister subpopulation remains a major challenge in this study and across the persister research field, as it requires isolating a pure persister population. While metabolic inhibitors like rifampin and tetracycline can induce dormancy and antibiotic tolerance in the entire population (Kwan et al., 2013), these treatments generate artificially altered cell states that may not accurately reflect naturally occurring persisters. Similarly, fluorescent reporters combined with fluorescence-activated cell sorting (FACS) have been employed to study persister cells, including in our previous studies (Amato et al., 2013; Orman & Brynildsen, 2013, 2015). However, this approach only results in persister-enriched populations rather than a pure isolate, meaning that persisters still constitute a small fraction of the sorted cells (Amato et al., 2013; Orman & Brynildsen, 2013, 2015). Despite these inherent limitations, our untargeted metabolomics and proteomics analyses at the whole-population level provide valuable insights into the regulatory mechanisms of the Crp/cAMP complex and its potential role in persister formation. Specifically, our data reveal clear indications that Crp/cAMP activity promotes the formation of a non-growing cell subpopulation, while its deletion reduces this effect. We have validated this observation through single-cell analyses (see Figure 4 in the current manuscript). Additionally, our data strongly suggest that energy metabolism plays a critical role in persister cell physiology, and we have rigorously tested this hypothesis using persister assays for single-gene deletions (see Figure 5 in the current manuscript).

Furthermore, in response to the reviewer's suggestion, we introduced *crp* and *crp* deletions into the HipA-7 high-persistence mutant strain. The impact of these deletions in HipA-7 mirrored their effects in the wild-type strain (Figure 1-figure supplement 8), further supporting our conclusions. This data has been provided and discussed in the manuscript (see lines 185-189, and Figure 1-figure supplement 8 in the current manuscript).

We acknowledge the challenges in directly assaying persister cells, and we have now discussed this in the manuscript (see lines 397-406 in the current manuscript).

(2) The authors overlooked/omitted a recently published work regarding cyaA and crp (PMID: 35648826). In that work, a deficiency in cyaA or crp confers tolerance to diverse types of lethal stressors, including all lethal antimicrobials tested. How a mutation conferring pan-tolerance to the major bacterial population would lead to a less protective effect with a minor subpopulation? The authors are kind of obligated to discuss such a paradox in the context of their work because that is the most relevant literature for the present work. It is also very interesting if the cyaA/crp deficiency really has an opposing effect on tolerance and persistence. As a note, most of the conclusions from the omics studies of the present work have been reached in that overlooked literature, which addresses mechanisms of tolerance, a major rather than a minor population behavior. That supports comment #1 above. The inability of the authors to observe tolerance phenotype with the cyaA or crp mutant possibly derived from extremely high antimicrobial concentrations used in the study prevents tolerance phenotype from being observed because tolerance is sensitive to antimicrobial concentration while persistence is not.

(3) The authors overly stressed the effect of *cyaA/crp* on persister formation but failed to test an alternative explanation of their effect on persister waking up after antimicrobial treatment. If the *cyaA/crp*-derived persisters are put into deeper sleep during antimicrobial treatment than wildtype-derived persisters, a 16-h recovery growth might have underestimated viable bacteria. This is often the case especially when extremely high concentrations of antimicrobials are used in performing persister assay. Thus, at least a longer incubation time (e.g. 48 and 72h) of agar plates for persister viable count needs to be performed to test such a scenario.

(4) The rationale for using extremely high drug concentrations to perform persister assay is unclear. There are 2 issues with using extremely high drug concentrations. First, when overly high concentrations are used, drug removal becomes difficult. For example, a two-time wash will not be able to bring drug concentration from $> 100 \times \text{MIC}$ to below MIC. This is especially problematic with aminoglycoside because drug removal by washing does not work well with this class of compound. Second, overly high concentrations of drug use may make killing so rapidly and severely that may mask the difference from being observed between mutants and the control wild-type strain. In such cases, you would need to kill over a wide range of drug concentrations to find the right window to show a difference. The gentamicin data in the present work is likely the case that needs to be carefully examined. The mutants and the wild-type strain have very different MICs for gentamicin, but a single absolute drug concentration rather than concentrations normalized to MIC was used. This is like to compare a 12-year-old with a 21-year-old to run a 100-meter dash, which is highly inappropriate.

The reviewer notes that key literature (PMID: 35648826) was overlooked, showing *cyaA/crp* deficiency confers broad stress tolerance—contradicting the reported reduction in persister protection. They suggest high drug concentrations may mask tolerance, and also, longer incubation (48–72 h) and normalized drug levels based on MIC are recommended. Given that these three independent comments are interconnected, we will address them together.

We follow a rigorous washing protocol to minimize antibiotic carryover. After treatment, 1 ml of culture is centrifuged at 13,300 RPM (17,000 $\times g$) for 3 minutes, and $>950 \mu\text{l}$ of supernatant is removed without disturbing the pellet. The pellet is resuspended in $950 \mu\text{l}$ PBS, diluting antibiotics >20 -fold. This step is repeated, resulting in a >400 -fold cumulative dilution. After the final wash, cells are resuspended in $100 \mu\text{l}$ PBS, then serially diluted and plated on antibiotic-free agar to ensure consistency and eliminate residual antibiotics. Preliminary experiments are routinely done in our laboratory to confirm the effectiveness of washing procedures. To address concerns that high antibiotic concentrations may mask phenotypic differences—particularly in the gentamicin assay—we conducted additional experiments using MIC-normalized doses ($5\times$, $10\times$, and the original study concentration) with six wash steps. As shown in Figure 1-figure supplement 6, all concentrations consistently reduced persister levels, supporting our original findings. While $5\times$ MIC ampicillin allowed detection of persisters in mutant strains, their levels remained multiple orders of magnitude lower than in wild-type, maintaining statistical significance. These results, along with updated washing protocols, are now included in the revised manuscript (see lines 176-185 and Figure 1-figure supplement 6 in the current manuscript).

Although we standardize the incubation time of the agar plates for all conditions and strains, most strains form sufficiently large colonies within 16 hours, and longer incubation often leads to large, overlapping colonies that hinder accurate counting. We assure the reviewer that we always leave the plates in the incubator beyond the initial counting period to monitor the emergence of any new colonies. Here, we provide plate images of key strains after antibiotic treatments, demonstrating that extended incubation did not alter CFU levels, as shown in Figure 1-figure supplement 7. We have updated the relevant section in the

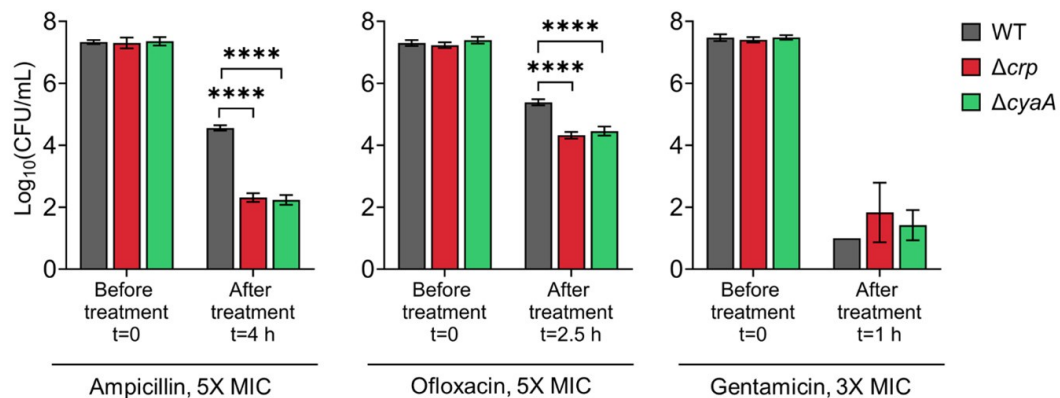
Materials and Methods to clarify this point and included the plate images in the current manuscript (see lines 181-182 and Figure 1-figure supplement 7 in the current manuscript).

We acknowledge the significance of the study highlighted by the reviewer (Zeng et al., 2022); however, direct comparisons with our results are challenging due to substantial differences in experimental conditions, antibiotic concentrations, treatment durations, and most importantly, the *E. coli* strains used. The study of Zeng et al., 2022, utilized strains from the Keio collection, a commercially available *E. coli* BW25113 mutant library, which may contain unknown background mutations that could influence tolerance phenotypes. While we used the Keio collection for initial screening, we always validate single clean deletions in our lab strain, *E. coli* MG1655, to ensure robust conclusions. The observed variations in tolerance and persistence between studies can largely be attributed to these methodological differences rather than an inherent paradox. The concentrations of ampicillin (200 µg/mL) and ofloxacin (5 µg/mL) used in our assays are in line with concentrations employed in foundational persister studies (Amato & Brynildsen, 2015; Cui et al., 2016; Hansen et al., 2008; Leszczynska et al., 2013; Lin et al., 2022; Orman & Brynildsen, 2015; Shah et al., 2006). These levels represent $>10 \times$ the MIC and are necessary to ensure the elimination of actively growing cells, thus enriching for persister cells that, by definition, survive high bactericidal drug exposure. Our aim is not to model pharmacokinetics per se, but to apply a standardized challenge to distinguish phenotypic persistence. Furthermore, pharmacokinetic and pharmacodynamic clinical data show that antibiotics such as ofloxacin and ampicillin can reach levels far exceeding $10 \times$ MIC for extended periods in patients (OFLOXACIN, 2019; Soto et al., 2014).

To assess how *cyaA* and *crp* deletions affect antibiotic responses under conditions similar to those used by Zeng et al. (Zeng et al., 2022)—specifically, exponential-phase *E. coli* BW25113 strains (Keio collection), lower antibiotic concentrations, and short treatments (e.g., 1 hour)—we first tested *E. coli* MG1655 WT, Δcrp , and Δcrp strains in late stationary phase using reduced antibiotic concentrations and shorter exposures. Both knockouts showed decreased survival following ampicillin and ofloxacin treatment compared to WT (see Figure 1-figure supplement 6), consistent with our findings in Figure 1 in the manuscript. In exponential phase, the knockout strains exhibited reduced survival after ampicillin treatment but increased survival after ofloxacin treatment relative to WT (see Author response image 2A below), again mirroring the trends in Figure 1. Gentamicin treatment, however, produced variable results in MG1655 knockouts, likely due to the brief 1-hour exposure being insufficient for robust conclusions (Author response image 2A). Notably, when we tested the corresponding Keio knockout strains in the BW25113 background, we observed increased tolerance in exponential-phase cells, reproducing Zeng et al.'s findings under their specific conditions (see Author response image 2B below), although BW25113 and MG1655 exhibited distinct persister phenotypes in exponential phase (Author response image 2A, B). These results, altogether, highlight the sensitivity of antibiotic tolerance and persistence phenotypes to factors such as strain background, antibiotic concentration, and treatment duration. This is now discussed in detail in the revised manuscript, with supporting data provided (see lines 460-476, and Supplement File 6, 7 in the current manuscript).

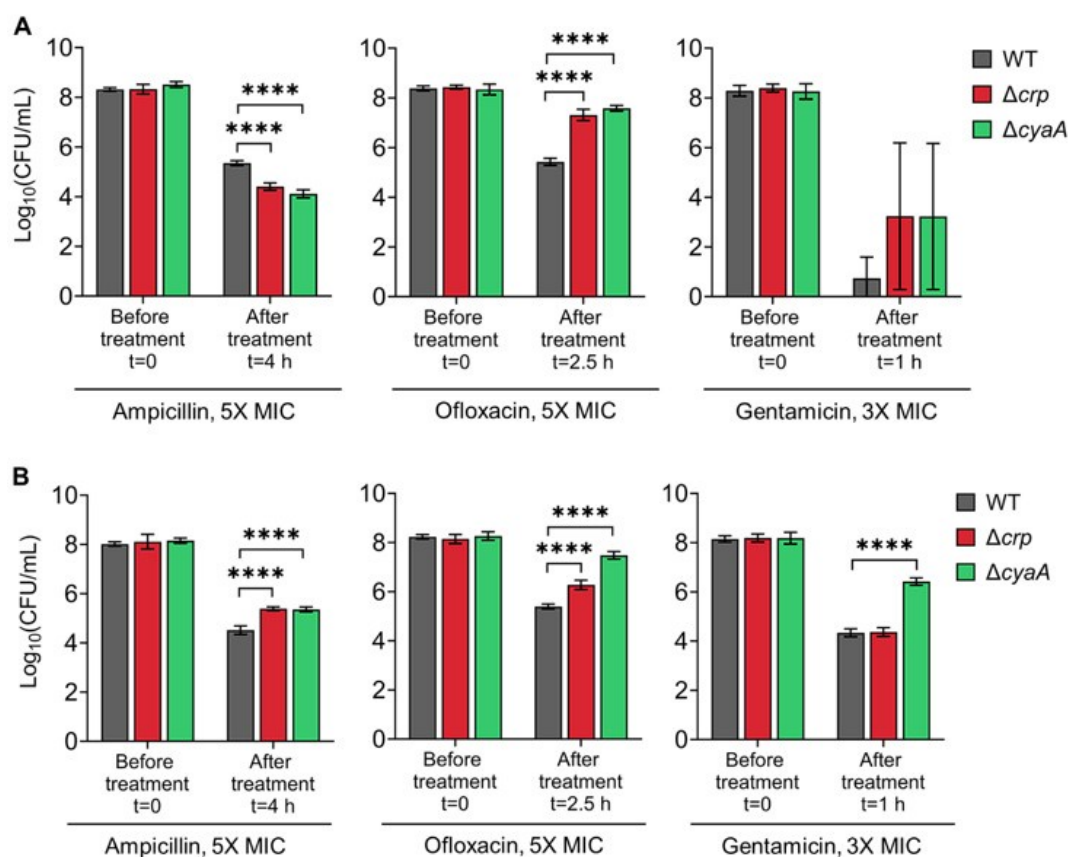
Author response image 1.

Persister levels of *E. coli* K-12 MG1655 WT, Δcrp , and Δcrp strains in late stationary phase. Cells were treated with ampicillin ($5 \times$ MIC for 4 h), ofloxacin ($5 \times$ MIC for 2.5 h), and gentamicin ($3 \times$ MIC for 1 h). Concentrations and treatment durations were selected based on (Zeng et al., 2022).



Author response image 2.

Persister levels of *E. coli* K-12 MG1655 (Panel A) and BW25113 (Panel B) WT, Δcrp , and $\Delta cyaA$ strains in the exponential growth phase. Cells were treated at mid-exponential phase ($OD_{600} \sim 0.25$) with ampicillin ($5 \times$ MIC for 4 h), ofloxacin ($5 \times$ MIC for 2.5 h), and gentamicin ($3 \times$ MIC for 1 h). Treatment concentrations and durations were based on conditions described in (Zeng et al., 2022).



Reviewer #3:

The authors try to draw too many conclusions and it's difficult to identify what their actual findings are. For instance, they do not have any interesting findings with

aminoglycosides but include the data and spend a lot of time discussing it, but it is really a distraction. The correlation between the induction of anabolic pathways in the crp mutant in the late stationary phase and the reduction in persisters is potentially very interesting but is buried in the paper with the vast quantities of data, and observations and conclusions that are often not well substantiated.

We thank the reviewer for their assessment that helped us clarify and strengthen the focus of our manuscript.

While our study is not focused on aminoglycosides, we believe the related data provide important insights into persister cell physiology. Persisters are traditionally described as metabolically dormant, non-growing cells. However, we consistently observe that aminoglycosides—despite requiring energy-dependent uptake and active protein translation for their activity—can still eliminate persister cells in wild-type *E. coli*. This finding supports our central hypothesis that persisters may retain a basal level of metabolic activity sufficient to permit aminoglycoside uptake and action during prolonged treatment. We have revised the manuscript to present this point more clearly, ensuring it complements rather than distracts from the main narrative.

We respectfully emphasize that our conclusions are supported by multiple layers of evidence. Our metabolomics data are corroborated by proteomics and further validated by functional assays, including redox state measurements, growing versus non-growing cell detection, and targeted persister assays. In addition, we performed labor-intensive validations using individually selected Keio mutants treated with antibiotics to quantify persister levels, with key observations further confirmed in single-gene deletions in *E. coli* MG1655 strains.

We believe the revisions made in response to all reviewers' comments have significantly improved the clarity, focus, and overall impact of the manuscript.

The discussion section is particularly difficult to read and I recommend a large overhaul to increase clarity. For instance, what are the authors trying to conclude in section (iii) of the discussion? That persisters in the stationary phase have higher energy than other cells? Is there data to support that? All sections are similarly lacking in clarity.

We repeatedly emphasize in the manuscript that while persister survival depends on energy metabolism, this does not imply that persisters have higher metabolic activity than those in the exponential growth phase. We have clarified this point in the revised manuscript (see lines 67-79, and 442-444 in the current manuscript).

The large number of mutants characterized is a strength, but the quality of the data provided for those experiments is poor. Did some of these mutants lose fitness in the deep stationary phase in the absence of antibiotics? Did some reach a far lower cfu/ml in the stationary phase? These details are important and without them, it is difficult to interpret the data.

Although metabolic mutations can affect cell growth, we do not observe substantial differences in cell numbers during the late stationary phase, when persister assays are performed. These knockout strains reach stationary phase fully by that time. We emphasize that we routinely measure cell numbers at this stage using flow cytometry before diluting cultures into fresh media and applying antibiotic treatments. Cell counts for the metabolic mutants are shown in Figure 5-figure supplement 4 in the current manuscript, and no significant growth deficiencies are observed in the late stationary phase. This is consistent with our previous publication (Shiraliyev & Orman, 2023) and findings from Lewis's group (Manuse et al., 2021), where similar knockout strains showed no drastic impact on growth.

There is an analysis of persister formation in mutants in the pts/CRP pathway that is not discussed (Zeng et al PNAS 2022, Parsons et al PNAS, 2024).

These studies are now cited and discussed in the revised manuscript (see lines 459-476).

The authors do not discuss ROS production and antibiotic killing in these experiments. Presumably, the WT would have a greater propensity to produce ROS in response to antibiotics than the crp mutant, but it survives better. Is ROS not involved in antibiotic killing in these conditions?

The experimental conditions used here are identical to those in our previously published study on persister cells in the late stationary phase (Orman & Brynildsen, 2015), where we specifically investigated the role of ROS in antibiotic tolerance. In that work, we overexpressed key antioxidant enzymes—catalases (*katE*, *katG*) and superoxide dismutases (*sodA*, *sodB* and *sodC*)—at stationary phase. These enzymes were confirmed to be catalytically active through functional assays, yet their overexpression had no measurable effect on persister levels. To further decouple ROS from respiratory activity in that study, we performed anaerobic experiments using nitrate as an alternative terminal electron acceptor. We found that anaerobic respiration actually enhanced persister formation, and inhibition of nitrate reductases using KCN reduced it—again, independent of ROS. These findings provide compelling evidence that it is the respiratory activity itself, rather than ROS production, that influences persister formation in our system.

We have now included this discussion in the revised manuscript to clarify that ROS are unlikely to be a major factor in antibiotic killing under these conditions (see lines 503-513).

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